



Stanwell New Project Office
Brian Hooper Architect/m3architecture
Façade Performance Specification

For Tender | Revision [00] | 19 December 2025

Document Control Record

Document prepared by:

Inhabit Australasia Pty Ltd

ABN: 77 136 869 942

Level 3, 545 Queen Street,
Brisbane, QLD 4000
Australia

T.+61 2 8003 6333

E contact@inhabitgroup.com

W www.inhabitgroup.com

Inhabit has prepared this document for the sole use of the Client named and recorded in Inhabit's Project Register + Accounting System + Email + Contractual records indicating professional services engagement. It takes into account client instructions and requirements. It is not intended for and should not be used by any other third party. Inhabit undertakes no duty, nor accepts any responsibility, to any third party who may rely upon or use this document. Verification of the validity of this document can be carried out by contacting Inhabit via its website at

www.inhabitgroup.com

Revision History

Report Title		Façade Performance Specification			
Rev	Date	Document Status/Details	Document Author	Reviewed by	Approved by
00	19/12/2025	For Tender	Cameron Gen	Kesa Kesawan	Kesa Kesawan

This document is to be read in conjunction with:

- **Façade Schedules** | 23909-SCH-FC0001[01]



Table of Contents

1.0	General	5
1.1	Introduction	5
1.2	Project Design Team	5
1.3	Work Included	5
1.4	Elements Not Included in the Works	6
1.5	Trade Coordination Responsibilities	6
1.6	Discrepancies	7
1.7	Warranties	7
1.8	Design Life	8
1.9	Relevant Experience	8
2.0	Submissions	9
2.1	Façade Schedules Requirements	9
2.2	Documentation	9
2.3	Manufacturers, Suppliers and Sub-Contractors	11
2.4	Engineering Design and Development	11
2.5	Quality Assurance	12
2.6	Certification	12
2.7	Inspections	12
2.8	Delivery, Handling & Storage	13
2.9	Project Conditions	13
2.10	Performance Testing	13
2.11	Spares	18
3.0	Performance and Functional Requirements	19
3.1	Façade Systems	19
3.2	Interfacing of Trade Packages	19
3.3	Façade Performance	20
3.4	Clip System for Structural Glazing Works	21
3.5	Structural Requirements	21
3.6	Connection Requirements	22
3.7	Movement Requirements	23
3.8	Environmental Requirements	25
3.9	Fire Performance	26
3.10	Maintenance Requirements	26
3.11	Visual Requirements	27
3.12	Tolerances	27
3.13	Safety in Design	28



4.0	Materials	29
4.1	Glass	29
4.2	Elastomeric Extrusions, Gaskets, Seals, Setting Blocks & Accessories	35
4.3	Structural Silicone	36
4.4	Weatherproofing Silicone	37
4.5	Thermal Barriers used in Metal Frames	38
4.6	Steel	38
4.7	ZINCALUME ^R Steel	46
4.8	Aluminium	46
4.9	Stainless Steel	50
4.10	Timber	50
4.11	Insulation	53
4.12	Prohibited Materials	53
5.0	Execution	55
5.1	Fabrication	55
5.2	Installation	68
5.3	Delivery, Handling & Storage	69
5.4	Project Conditions	69
5.5	Completion	71
A	Appendix A - Reference Standards	72
A.1	Loadings	72
A.2	Glass	73
A.3	Glass Coatings	73
A.4	Glass – Heat Treated	73
A.5	Glass – Laminated	74
A.6	Glass – Insulating Glass Units	74
A.7	Glass – Bent/Curved	74
A.8	Aluminium and Framework	75
A.9	Concrete & Masonry	75
A.10	Anchor Fixings	75
A.11	Steel	76
A.12	Stainless Steel	76
A.13	Zinc	77
A.14	Timber	77
A.15	Glazing Framework	77
A.16	Sealant	77
A.17	Gaskets	78
A.18	Coatings	79



A.19	Cladding Systems and Waterproofing	79
A.20	Access & Maintenance	79
A.21	Testing	80
A.22	Thermal Calculations	80
A.23	Hardware	81
A.24	Other	81
B	Appendix B – Contractor Submissions	82
B.1	Design and verification submissions	82
B.2	Fabrication submissions	84
B.3	Installation submissions	88



1.0 General

1.1 Introduction

This document is a performance specification forming part of a design and construction package, for the façade and associated works.

This Specification is to be read in conjunction with the **Façade Schedules 23909-SCH-FC0001(01)**.

This Specification (and the **Façade Schedules**) is to be read in association with all other contract documents related to these works. Unless otherwise stated, all obligations for these works are to be undertaken by the Façade Contractor.

Refer Section 1.0 of the **Façade Schedules** for a list of the Contract Documents.

1.2 Project Design Team

Refer Section 2.0 of the **Façade Schedules** for the project design team.

Unless noted otherwise the term “approved or approval” refers to approval by the Client or the Client’s nominated representative.

1.3 Work Included

It is recognized that the Contract Documents may not cover some conditions or accurately reflect some proposed conditions. It is intended that the conditions not detailed or drawn accurately shall be developed through the Façade Contractor’s shop drawings to the same level of aesthetics and performance criteria as indicated for detailed areas and as stipulated in the Contract Documents. The Façade Contractor represents that he has, in the preparation of his proposal, visited the site and determined existing conditions. By accepting a contract for the Work, the Façade Contractor acknowledges the above and agrees that the Client or the Client’s representative shall have final decision as to all matters whether detailed or not in the Contract Documents.

The Façade Contractor shall design, engineer, test, fabricate, deliver, install, and guarantee all construction necessary to provide for the Façade Systems for the watertight enclosure of the building. The façade shall be complete in every respect, including all measures that may be required to that end, notwithstanding any omissions or inadequacies of Drawings and/or Specifications. The Work of this Section shall include, but is not limited to the following:

- Façade systems as described in the **Façade Schedules**.
- Entrances and operable doors (including panel lift doors), including all hardware, and operational controls required for a complete and operable assembly.
- Miscellaneous steel support framing and architecturally exposed structural steel, unless specifically excluded in the **Façade Schedules**.
- Parapets, copings, gutters, soffits and backpans attached to the façade.
- Louvres, railings, flashing and closures between floors occurring in the façade.
- End closures at all horizontal and vertical caps and projections to the façade.
- All anchors, fixings, attachments, and reinforcements except those specifically indicated as being provided by other trades.
- All smoke barrier, waterproofing and fire safing attached to or within the façade system including supports, backing, and reinforcements.



- All gaskets, sealants, elastomeric and metal flashing including sealing at junctions with roofing, ground floor waterproofing, and building expansion joints.
- Finishes, protective coatings and treatments.
- Stillages, lifting beams, and other installation equipment required for installation of the Façade Systems
- Electrical bonding and earthing of all metal cladding and other miscellaneous components describe above.
- System penetrations and seals for plumbing services including fire sprinkler soak system, spitters and overflows.
- Fixing brackets for the attachment of BMU restraint sockets that are to be provided by the BMU contractor, where possible these are to be factory fitted.
- Provisions for electrical outlets and cut-outs for lighting and security, conduits and other electrical work that are integrated into the façade.
- All internal aluminium sills, trims, pelmets and reveals to windows.

It is recognized that the Façade Documents may not cover all conditions, so it is intended that the conditions not detailed or drawn accurately shall be developed through the Façade Contractor's shop drawings to the same level of aesthetics and performance criteria as indicated for detailed areas and as stipulated in the Contract Documents.

1.4 Elements Not Included in the Works

The following elements shall be excluded from the works:

- Internal steel framed walls and block work walls including their structural reinforcement, thermal insulation, water and vapour barriers, and internal finishes. These walls are provided by others and are required to be structurally sufficient to support the external façade system.
- Primary concrete and steel structure
- Ceilings, floors and floor finishes
- Blinds, blind fixing brackets and rails
- Mechanical, electrical and hydraulic systems
- Civil, paving, landscape or other external works

1.5 Trade Coordination Responsibilities

The Façade Contractor will be required to collaborate with the local project delivery team to coordinate the design and engineering of the Works, including but not limited to the following:

- Primary building and secondary structural framing, including loading and set-out, cast-in components
- Other façade trade contractor works, including interface details
- Roofing trade contractor works
- Building services – including mechanical, hydraulic, electrical, lighting and lightning protection
- Internal finishes – including floor finishes, ceilings, internal linings, partitions and blind provisions
- ESD provision compliance
- Building management systems
- Waterproofing membrane works
- Fire services
- Security
- Signage provisions
- Building access strategy and provisions – Cleaning and long-term maintenance



It is to be understood by all parties that all systems described in this specification and accompanying drawings are to be designed, engineered, tested, fabricated, delivered, installed and guaranteed by the Façade Contractor as part of a 'Design and Construct' contract agreement.

1.6 Discrepancies

All work shall comply with the requirements of the Contract. In any instance where there is a discrepancy, agreed to in writing prior to commencement, it shall be amended and or as soon as the non-compliance is identified. Where there is a contradiction in terms, the more onerous condition applies.

Should any discrepancy or divergence occur in or between any of the Contract Documents as to the precise extent or nature of the work to be carried out by the Façade Contractor, they shall immediately give notice to the Main Contractor so that the Main Contractor's direction or instruction may be given as to the work required by the Façade Contractor.

1.7 Warranties

Refer to the **Façade Schedules** for warranty periods.

1.7.1 Warranty Conditions for Façade Systems

The warranty shall include specific provisions and terms for failure of, or due to, the following:

- Failure of system to exclude water and weather or deliver water to drainage system.
- Structural integrity, including dislodgement due to wind within the design load range.
- Damage, deterioration, perforation or staining due to contact or close proximity with dissimilar metals or chemical products in the substrates, adjacent materials or the environment generally.

1.7.2 Warranty Conditions for Sealants

The warranty shall include provisions and terms for failure of, or due to the following:

- Cohesion and adhesion.
- Staining or bleeding to adjacent surfaces.
- Change in hardness greater than ± 5 points on the Shore A durometer scale.
- Fading or colour change (greater than 2 Rating Scale points measured in accordance with AS 1580.481.1.2, Table 1 or greater than 5 Hunter Colour Difference Units measured in accordance with ASTM D2244 for products of US origin), or colour variation in adjacent sections greater than the variation of the approved control sample range.

1.7.3 Warranty Conditions for Applied Coatings

The warranty shall include specific provisions and terms for failure of, or due to, the following:

- Failure of protective coating to prevent corrosion and staining due to corrosion.
- Suitability of coating system, including preparation and base-coats, for substrate types.
- Delamination, peeling, flaking, chipping, drumminess, loss of bond to substrate.
- Cracking, blistering, perforations or similar defects (visible from 500 mm).
- Ultra-violet light, atmospheric pollution or weathering.
- Staining and discoloration.
- Chalking (greater than 8 points for non-metallic colours or 6 points for non-metallic whites, measured in accordance with ASTM D4214).



- Fading or colour change (greater than 2 Rating Scale points measured in accordance with AS 1580.481.1.2, Table 1 or greater than 5 Hunter Colour Difference Units measured in accordance with ASTM D2244 for products of US origin), or colour variation in adjacent sections greater than the variation of the approved control sample range.

1.8 Design Life

This is the Duration of the period during which a structure or a structural element, when designed, is assumed to perform for its intended purpose with expected maintenance but without major structural repair being necessary, as defined by AS1170.0. Refer to the **Façade Schedules**.

1.9 Relevant Experience

The Façade Contractor shall provide details of existing examples in Australia of the proposed or similar systems, including name, address and contact details of an independent person knowledgeable about each example.

Submit details of experience of all proposed subcontractors and suppliers demonstrating continuous experience with the proposed systems and/or specified products and their required roles.

Proposals to use suppliers or personnel with experience less than the periods indicated must include reasons why the lesser experience is considered acceptable by the Main Contractor. These proposals may be considered by the Client, but will be subject to their approval.

Façade Contractor	Not less than 10 years
Façade System Designer/s	Not less than 10 years
Façade System Engineer/s	Chartered Engineer with no less than 10 years' experience and where applicable must also have building practitioners license applicable to that Australian state/territory.
Installer	Not less than 5 years
Glass Manufacturer	Not less than 15 years
Glazier	Not less than 5 years
Sealant Manufacturer	Not less than 15 years
Supervisor and Trade Quality Coordinator (TQC)	Not less than 10 years
Independent Testing Authority	NATA, HOKLAS or SINGLAS approved (or other approved ILAC MRA signatory)



2.0 Submissions

2.1 Façade Schedules Requirements

This specification is to be read in conjunction with the project **Façade Schedules**, which includes among other things:

- Proposed Project Team
- Tender Submissions
- Detailed Program (covering submissions, testing, production, hold points)
- Proposed Meetings Schedules
- Subcontractor – Factory (supplier)
- Subcontractor – installer
- Materials Submissions
- Shop drawing and engineering submissions
- System tolerances and movement limits
- System thermal performance U values
- System acoustic performance verification and testing where necessary
- Surveys to be co-ordinated with Main Contractor
- Interface co-ordination details (adjoining trade packages)
- Visual mock-ups
- Performance Test Proposals
- Details and reports of tested systems (including drawings)
- Toughened glass heat soak test records
- Warranties
- Hold Points
- Quality Plans
- As-Built Records
- Spares
- Operation and Maintenance Manuals
- Temporary protection
- Builders clean
- Final clean

2.2 Documentation

Provide all required submissions for a comprehensive performance proposal, to demonstrate compliance with the required design intentions and performance requirements, record construction methods and procedures, record materials and components proposed, and to establish quality standards. Refer to the **Façade Schedules** for the full list of submissions.

2.2.1 Computations

Submit engineering computations indicating structural sufficiency, safety and performance.

Computations shall include a system summary, design methodology, loading assumptions, material properties and list of performance criteria including those requirements specified in the **Façade Schedules**.

Drawing numbers and details are to be noted to which the computation relates.

Include Engineering Computations for:



- Dead loads, live loads (including maintenance and service loads) and where applicable earthquake and snow loads
- Human impact and safety requirements
- Movements, deflections and tolerances and provide clarification as to how these are accommodated by the system
- Framing members, stiffeners and reinforcement
- Glass types and thicknesses
- Cladding panels and stiffeners
- Hardware including locking point positions e.g. operable sash windows
- Connections, anchorages, brackets, fasteners and fixing to the primary structure
- Secondary framing, brackets and anchorages to the primary structure

Provide additional tests of all fixings and connections which have not been calculated, to demonstrate ability to carry the applied loads adequately. Post-installed and cast-in fastenings in concrete must comply AS 5216.

Engineering computations shall be prepared and certified by a Chartered Professional Engineer with no less than 10 years' experience and where applicable must also hold a building practitioner's license applicable to local state/territory. Submit details and experience of proposed engineer.

2.2.2 Shop drawings

Submit comprehensive, detailed and dimensioned shop drawings to indicate all set out and construction details not indicated in the Contract Documents, adjacent work by others, and for all proprietary products where required details are not indicated in the manufacturer's product data.

Do not commence manufacture until approval to use the relevant shop drawings has been obtained.

Include and indicate:

- Shop drawing register
- Material and/or component schedule included
- Set out of all work, including reference points, edge conditions and joint pattern, indicated on plans, elevations and sections as applicable.
- Full size sections of all members, including descriptions of structural properties and specifications of materials.
- Framing, anchorages and fixings supported from base-structure, and embedments in the base-structure, where required.
- Movement joints.
- Methods of assembly at all junctions, including sealing and fixing, indicated by three-dimensional and exploded views in three dimensions if requested.
- Method of installation, including:
- Erection tolerances.
- Machined slots, keyholes and other methods for handling and connecting components.
- Junctions and trim to base-structure and adjoining surfaces.
- Fully dimensioned set out drawings and templates.
- Access and materials handling equipment and requirements.
- Hardware, fittings, accessories and dimensional constraints.
- Methods of meeting performance criteria for thermal insulation, fire resistance, sound transmission and the like.
- Method of cleaning in service.
- Panelling details, including all joints and junctions, suspension systems and panel reinforcement.



- Connection details and provision of lightning protection.
- Record Shop Drawings
- Interfaces with and terminations against neighbouring façade systems using coordinated documentation that is accurate for construction details

2.2.3 Prototypes

Refer to the **Façade Schedules** for a full list of the required prototype submissions.

2.2.4 Maintenance and Operating Manual

Submit a comprehensive manual containing routine for operating and routing cleaning and maintenance, and all information required to ensure the full-service capability of the work, including source of replacement components and methods of replacement of damaged components for a period not less than the warranty period. This is to be complete prior to the completion of the work.

The method of cleaning and timetables are to be specified together with cleaning agents which can and cannot be used.

2.2.5 As Built drawings

Upon completion of the façade, the Façade Contractor shall prepare and endorse the necessary As Built drawings for submission to the architect.

2.3 Manufacturers, Suppliers and Sub-Contractors

The Works described in this Specification shall be the responsibility of one Façade Contractor.

The Façade Contractor of the Work described in this Specification shall have proven achievement and experience with façade systems as specified, and are subject to final approval by the Main Contractor/Architect.

Manufacturers and suppliers of all materials and components of the Work described in this Specification are subject to approval by the Main Contractor/Architect.

2.4 Engineering Design and Development

2.4.1 Engineering

Engineering design development, materials and methods of construction other than that indicated or implied by the Contract Documents may be employed when such materials and methods conform to all of the following:

- The design intent
- The performance criteria (Strength, Durability including colour retention, Weather-tightness, Fire Resistance, Acoustic and Thermal Performance)
- All applicable codes and standards
- Approval by the Main Contractor and Client or Client's Nominated Representative

2.4.2 Façade Contractor's Professional Engineer

The Façade Contractor shall retain an experienced Professional Façade Engineer. Prior to commencement of Work, the Façade Contractor shall submit the Professional Façade Engineer's credentials for approval.



- The Façade Contractor's Professional Façade Engineer shall prepare and endorse complete engineering design and calculations and shall check and monitor the preparation of all shop drawings for conformance with the engineering design and calculations, and for compliance with the Contract Documents.
- As evidence of conformance to these requirements, each shop drawing shall bear the seal and self-written signature of the Façade Contractor's Professional Façade Engineer verifying that it has been examined and approved.
- The Professional Façade Engineer shall attend site during installation of the façade and confirm that installation is in accordance with the engineer's certified design.

2.4.3 Timing

- Computations are to be prepared and made available for review in conjunction with the shop drawings.
- These computations are to be submitted at the same time as the relevant shop drawings and component schedule.

2.5 Quality Assurance

Prepare, submit, and implement a detailed project specific Quality Plans for the following.

- Float glass manufacture
- Glass processing (all steps)
- Extrusion factory
- Paint and anodizing
- Metal fabrication and coating
- Timber production and fixings
- Fabrication / assembly
- Hazardous materials checking
- Packing and Transportation
- Installation

2.6 Certification

Submit certification as follows:

- Performance requirements - Compliance of performance proposals with design intentions indicated on the Drawings, specified requirements, and relevant statutory and project requirements before commencing fabrication
- Substrates – Acceptance of suitability of substrates before commencing installation
- Completed work - Compliance of all components, materials and completed work with the Contract Documents and all performance requirements at completion of installation.

2.7 Inspections

Give sufficient notice (not less than 24 hours) so that inspection may be made of the following:

- Required prototypes constructed and ready for inspection or testing
- Commencement of any required testing on or off site
- Fabricated assemblies at the factory ready for delivery
- Building substrates, including sub-framing, prepared and ready for the installation of the fabricated assemblies, with anchor brackets and other attachments fixed in place



- Initial set-out of sub-framing and bracket installation of each system
- First occurrence of sub-framing and bracket installation of each system
- First occurrence of installation of finished work of each system (first 20 m2)
- Commencement of final capping/sealing of the building

For panel inspections in the factory, the minimum number of panels to be available for inspection shall be agreed with the Architect, but generally shall be in logical groupings of finished work, arranged to allow all faces, returns and edges to be viewed and compared with approved samples under daylight conditions for both close and comparative inspection.

2.8 Delivery, Handling & Storage

All materials shall be packaged and palletted for safe and convenient handling by fork-lift vehicles and the Main Contractor's crane and hoisting equipment being used on the site.

The Façade Contractor is to ensure that all stillages, lifting beams, and other installation equipment is fully engineered, load limits are clearly stated and this equipment is inspected in accordance with Worksafe and other statutory requirements before this equipment is able to be used on the project.

Provide all required packing, wrapping and handling techniques to ensure that all panels are installed and handed over without any damage to the finished panel faces. Any damage to surface finishes may be ground for rejection and replacement.

Prefabricated panels stored in the factory or on site shall be stored under cover and not exposed to the weather. In the event that prefabricated panels cannot be stored under cover, they shall be stored in their stillages and shall be wrapped to prevent water ingress.

2.9 Project Conditions

2.9.1 Survey and Set-Out

Allow for engaging an experienced licensed surveyor to progressively verify actual locations and tolerances of profiles, openings, embedments and defects in base-structure, and make adjustments before fabrication and installation. Also address building movements that occur during construction in setting out the façade works.

2.10 Performance Testing

2.10.1 Performance Mock Up (PMU) / Prototype

2.10.2 General

The Façade Contractor shall allow for a full-scale façade prototype to be performance tested in a registered laboratory or test facility. The test shall be carried out in accordance with the testing standard as nominated in the **Façade Schedules** e.g. AS/NZS 4284, AS 2047 or an equivalent test standard approved by the design team.

The Façade Contractor shall:

- Carry out and pay for testing by an approved registered testing laboratory of a full-scale façade system prototype and provide test results to demonstrate compliance with the performance requirements in accordance with the nominated testing standard.
- Test prototype, including a suitable simulated base-structure frame, shall be erected off-site.
- Carry out testing of façade system. Each Trade Section shall coordinate relevant aspects of the test and cooperate as required.



- Provide accurate shop drawings for the proposed prototype. Do not commence construction of prototype until approved in writing.
- Modify prototype as required to comply with design intent, and re-test until full compliance is achieved.
- Submit test results progressively.

All costs associated with the performance test shall be allowed for by the Façade Contractor. This includes costs associated with fabrication, assembly, dis-assembly, transport costs to and from the site, and all fees and charges associated with the test facility. Test Laboratory and Testing Program.

2.10.2.1 TESTING FACILITY

The test facility shall be an independent registered laboratory that has obtained ISO/IEC 17025 accreditation. Details of the intended test facility including curricula vitae of the relevant testing personnel and an accreditation certificate no more than 6 months old, shall be submitted for review and approval.

In addition to ISO/IEC17025 requirements, the testing laboratory shall be accredited to undertake all aspects of the nominated testing, and a NATA, HOKLAS or SINGLAS accredited laboratory. Other mutually recognised countries may be considered subject to review and approval. Should the test facility be located outside of Australia, the Façade Contractor shall include provision for a witness from the Client, Superintendent and the Architect to be present at the prototype test.

Contact details at test facility are to be provided to permit verification of testing, testing methodologies and results if required.

2.10.2.2 TESTING PROCEDURE

The Façade Contractor shall submit a detailed and sequential testing program with diagrams indicating the type and location of data collection instruments. Do not commence testing until approved.

Submit notice in writing of proposed commencement of the performance testing program not less than one week before commencement.

No pretesting of the test assembly shall be allowed except by express written consent from the Architect, Superintendent or the Client.

The test assembly shall be erected under supervision of the same personnel that will supervise installation of the façade at the project site.

Façade Contractor shall submit for approval detailed descriptions and shop drawings of test assembly, transducer locations, test procedures, test schedule, test location, and test report consistent with the submittal requirements of the Contract Documents.

Façade Contractor shall submit a digital recordings throughout the construction and testing of the test assembly. Record the activities using equipment capable of high quality recording at short play speed or equivalent resolution settings.

- Record the installation of each individual component or trades work incorporated into the test assembly. No deviations from the approved shop drawings shall be allowed unless otherwise approved.
- Record the testing program including any pre-testing and carefully note all failures and remedial action undertaken.
- Record the replacement procedures for the removal and re-installation of selected aluminium components.



Test reports shall include:

- The original report issued by the laboratory, signed by the testing officer in charge, including certification that the testing has been carried out in accordance with the relevant testing codes and the Contract Documents.
- Detailed 'prototype as tested' shop drawings, indicating actual measured dimensions of component parts and remedial work carried out since previous tests.

2.10.2.3 SUBMISSIONS

Prior to the commencement of the test, the Façade Contractor shall submit a proposal outlining the method of test, design loads and transducer locations for review and approval. The test proposal must be accompanied by accurate shop drawings; failure to comply will result in the test proposal being rejected.

Submit all test results, including preliminary and interim test results. Indicate results in accordance with 'acceptable performance criteria'. Content should be formulated to match the nominated testing standard requirements e.g. AS/NZS 4284, Appendix A Specific Test Requirements. Refer also to the **Façade Schedules** for submission and load test requirements.

Pass/fail results shall comply with the Contract Documents or relevant codes whichever is the more stringent. In addition:

- There shall be no permanent deflection of any member.
- There shall be no leaks from any water penetration control devices.
- Water which enters the façade system shall drain to the exterior within one minute of ceasing the positive pressure differential.

2.10.2.4 DESIGN ELEMENTS TO BE TESTED

Test prototypes shall incorporate at least one example of each repetitive design element and construction method in the system, including all configurations, sealants, adhesives and gaskets.

The prototypes shall include:

- Attachments to the structure.
- Each type and subtype of frame member.
- Lifting eyes and inserts for installation where applicable.
- Panel junctions.
- Horizontal and vertical movement joints.
- Structural and other sealants, gaskets, tapes and beads.
- Junctions and trim to adjoining surfaces.
- Caulking and flashing.
- Hardware, fittings and accessories.
- Minimum test size of performance mock-up is to be a minimum 4 panels wide x 1 panel high, unless otherwise approved.

2.10.2.5 PREVIOUSLY TESTED SYSTEM

In the instances where an identical system has been previously tested within the past 5 years to higher loads, then it may be considered in lieu of testing provided that testing was undertaken by a NATA, HOKLAS or SINGLAS accredited laboratory accredited laboratory (or other approved ILAC MRA signatory), and the test data and PMU shop drawings are submitted for verification, review and approval.



For a previously tested system to be considered, the Façade Contractor shall submit for review and approval:

- Shop drawings of previously tested system showing use of the same extrusions, material grades, fixings and fabrication. Previously tested system must be of a similar geometry
- Test Report including pass/fail rates
- Test Report showing loads tested to. Loads must be equal or higher than the performance requirements for this project
- The test prototype must have been supported on a structure which simulates the structural behaviour, materials, profiles, dimensions and finishes for this project
- Certificate confirming that testing laboratory was ISO/IEC 17025 accredited and was approved to undertake the nominated test e.g. AS/NZS 4284.

2.10.2.6 TEST PROTOTYPE AND TEST CHAMBER

The test prototype shall be identical to the proposed façade system materials, fabrication and finishes and match previously approved samples. Interior finishes are not required.

Support the test prototype on a structure which simulates the structural behaviour, materials, profiles, dimensions and finishes of the actual on-site base-structure in all important respects. Include infill panelling required to seal the prototype to the test apparatus.

Prototype shall be square and plumb, complete with weather seals, fixings, trims and flashings.

An externally mounted chamber of a size to fit the test prototype shall be provided. The fit shall be such as to seal the perimeter of the test prototype against air and water penetration.

2.10.3 Field Testing

2.10.3.1 FLASHING AND GUTTER TESTING

Conduct a complete gutter test to the first completed floor as prior to proceeding with the next floor and then carry out 10% gutter testing unless otherwise agreed. Gutter testing is to be carried out by temporary blocking gutters and filling them with water. Gutters must be unblocked prior to proceeding with panel installation.

2.10.3.2 HOSE TESTING

Conduct hose testing to the first 25sqm of each façade type prior to proceeding with full installation and then carry out additional 25sqm hose testing every 250sqm unless otherwise agreed.

Allow for field water testing to be carried out periodically during façade system installation to AAMA 501 for all face sealed systems. The location and frequency for hose testing shall be as per the **Façade Schedules**.

2.10.4 Structural Silicone Testing

The following periodic testing of structural silicone shall be carried out, documented, and submitted for review:

- Peel Test
- De-glazing Test
- Migration Test
- Snap Test
- Butterfly Test



Obtain from the specialist silicone supplier and submit for review their procedures for fabricating structural silicone glazed units (glass unit or cladding panel bonded by silicone onto a subframe).

Provide a letter from structural silicone supplier stating that they have received design of structural silicone joints, witnessed fabrication, QA testing, audited QA records and procedures. Based on the above they warrant the design and fabrication of the structural silicone.

Provide the following:

- Silicone batch log to record all batches used including batch manufacture, date and arrival date of each batch at the fabrication works.
- Frame logging at time of assembly, which shall include identifying every panel by a unique number. The glazing record for each panel shall include silicone type, batch reference and date, curing time, date of application, glazier's name, and temperature and humidity measured inside the factory at a nominated time of each day.
- Upon request provide the following:
- Documentation of the sealant manufacturer's requirements for the particular substrates of the construction including joint sizes, limitations, requirements for mixing, cleaning, surface preparation, priming and application.
- Glazing procedures including frame assembly, cleaning, priming (if necessary), gunning, tooling, frame handling after glazing and curing.
- Details of testing of silicone being used on a weekly or batch basis whichever is the least.
- Details of tensiometer and any other testing equipment as required.
- Details of substrate testing carried out on a daily basis to ensure continued high quality of and consistency of silicone adhesion to the substrate.
- Details of de-glazing. Choose frames at random as follows:
 - 1 out of the first 10 frames (frames 1 to 10)
 - 1 out of the next 40 frames (frames 11 to 50)
 - 1 out of the next 50 frames (frames 51 to 100)
 - 1 out of each 100 frames for the remainder of the project.
- Details of an acceptance criteria for the periodic de-glazing
- An assessment of the quality of the de-glazed units against the requirements for cohesion, adhesion, tear resistance and compatibility noted herein.
- Details of procedures to deal with non-conformities. The procedure shall provide for the identification of all frames of unacceptable quality and their re-glaze or rejection. When non-conformity is established as a result of a random de-glaze, then the independent inspection authority shall call for further de-glazing of the frames glazed on the same day and using the same silicone batch as the rejected frame.
- Compatibility test certificates by the structural sealant manufacturer: Test for compatibility of the sealant used with all substrate materials including aluminium, finishes, glass, glass coatings, gaskets, setting blocks, backing rods etc.
- Adhesion test certificates by the structural sealant manufacturer: Carry out tension and shear tests to rupture. Note whether cohesive or adhesive failure occurred in the tension rupture tests.

2.10.5 Manufacturer's Standard Test Reports

Submit manufacturer's current test reports verifying physical characteristics of proposed systems and components. Reports shall be prepared by a registered testing laboratory and shall include details of test standards and dates of testing.



If directed, submit copies of the test standards used.

Where there are any differences between the items tested and the items proposed for use in this Trade Section, submit additional information to describe such differences, and carry out additional testing if directed.

2.10.6 Embedded Anchor Testing

If cast in embedment anchorages exceed product specification, the following is to be provided.

- Representatives of each anchor type embedded into concrete or reinforced masonry shall be site tested.
- Each anchor type shall be tested for shear, tension, and shear and tension combined to 2.0 times the design load.
- Testing shall comply with ASTM E 488, Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements, or BS 5080 Part 1:1993.
- A minimum of 5 per cent but no less than 5 anchors of each anchor type shall be tested.
- There shall be no failure or permanent deformation to the anchor or the concrete or reinforced masonry.

2.11 Spares

Main contractor to confirm allowance with Client at the time of Tender and is included in the Contract Documents. At or before practical completion, deliver to site in strong protective packages marked for identification and store where directed, spare components for future replacement or repairs



3.0 Performance and Functional Requirements

3.1 Façade Systems

3.1.1 Unitised Façade Systems

Unitised façade systems e.g. Curtain wall, window wall are required to be unitised, pressure equalised systems with continuous front rain-screen legs and continuous rear air-seal legs. Profiles are to be designed to accommodate the full range of building movements and are to be drained to prevent standing water being captured inside the façade system.

3.1.2 Rain Screen Cladding

Rain-screen cladding is to be open jointed or sealed as nominated by the Architect. There are to be no exposed or visible fixing at the panel joints. All rain-screen cavities are to be vented and drained. They are to be flashed off every 10m both vertically and horizontally with aluminium or stainless-steel flashing. All components are to be fully engineered.

3.1.3 Shop Front Glazing

Shop front glazing systems are to be provided in accordance with the Architect's drawings with particular attention given to providing visual quality and weather-tightness. Where face sealed systems are used, sealants, tapes and gaskets are to be carefully aligned to provide straight, uniform appearance with tolerances not exceeding +/-1mm per metre.

3.1.4 Operable Windows

Include a project-specific assessment of risks associated with operable windows in the Safety in Design assessment. Refer to clause 3.13 Safety in Design.

Operable windows are to incorporate:

- Multi-point locks with self-latching mechanisms
- Lockable handles
- Child resistant release mechanism if the screen or device can be removed, unlocked or overridden in accordance with NCC requirements.
- Finger entrapment warning sticker

All operable windows shall be provided with engraved metal tags to be permanently fixed to the sashes warning that the window is to be kept closed during strong winds.

Awning sash windows are to be top hung via a continuous hook detail with adaptor to prevent disengagement.

3.2 Interfacing of Trade Packages

The design, fabrication and installation of the Works are to be fully coordinated with all interfacing elements including, but not limited to:

- Façade, roofing, and drainage systems provided by others.
- Adjacent finishes
- Ground level and/or terraces and balconies
- Internal fit out/linings



Clear responsibility is to be documented at all interfaces between trade packages and is the responsibility of the Main Contractor to coordinate and communicate the design and documentation responsibility.

Below are design principles to be considered at typical interfaces –

- Where interfaces exist between systems the principle of two steps and two seals is to be incorporated, with cavities and gutters allowing for a clear drainage path to the exterior of the façade systems.
- Allow for all flashings (primary & secondary) at parapet interface to roof decks and terraces.
- Design flashings to allow for maintenance loads, and movements, as applicable.
- Sill details to incorporate freeboard requirements in accordance with AS4654 Part 1 and Part 2, and project wind speed requirements.
- Provision of access requirements (rope guide rails or similar) and their connection beyond and interface to external weather lines is to be considered.

3.3 Façade Performance

Individual and aggregate components of the façade system shall be designed, fabricated, assembled, transported, installed and protected so that no evidence of the following shall be apparent, visually or measurably, when Work of this Specification is subject to the pressures, loads, temperatures, rain, hail, snow where relevant and conditions specified:

- Damage of any kind
- Permanent deformation
- The service life of the façade system shall be not less than that specified in the **Façade Schedules**.

3.3.1 Corrosion Protection

Provide protective coatings to prevent corrosion of metal components, including concealed and structural components.

All fixings required for the attachment and support of the system to the base structure shall be permanently resistant to corrosion and completely protected from the weather.

3.3.2 Compatible Materials

Adjacent materials and products generally shall be chemically and electrolytically compatible with each other and with relevant substrates and shall be non-staining and non-bleeding.

Metals and other dissimilar or incompatible materials shall be isolated by approved coatings, spacers and other means to prevent corrosion, perforation of protective coatings, or other damage due to electrolytic action, chemical attack, or by any other cause including bad practice, incorrect installation, or incorrect fixing methods.

Indicate on shop drawings provision for separating dissimilar materials including concrete to aluminium. Include treatment of items embedded in concrete or grouted over.

Where there is a known or anticipated incompatibility between materials proposed in this Specification, and/or materials proposed in this Specification and adjacent materials supplied or proposed under other Trade Specifications, submit complete analysis of compatibility issues, including selection of materials and metals in close proximity, and details of methods of proposed separation, coatings and spacers.

Submit certificates from relevant manufacturers confirming acceptance of the use of the selected products in the manner and on the substrates proposed.



3.3.3 Dimensional Stability

The system shall have adequate dimensional stability to function properly and prevent damage to adjacent or applied work by others.

3.3.4 Safety and Human Impact

All glass for balustrades, glass walls and the like shall be designed to resist human impact loads in accordance with the relevant Standards.

Computations and shop drawings demonstrating compliance are required to be submitted.

No monolithic toughened glass is permitted for use in any lite where the fall to finished floor level below is over 3000mm in the event of glass breakage, unless otherwise agreed with design team.

3.3.5 Panel Replacement

Submit details of replacement procedures, that addresses all glass on the building, temporary safety measures, clip or other fixing details and equipment required. Include estimated maximum lead time required by the glass manufacturer to supply replacement units for each type of glass unit.

3.4 Clip System for Structural Glazing Works

Integrate the provision for a clip in restraint system to the glass in instances where four-sided structural silicone glazing is used. The clip system is to be designed for both temporary restraint during re-glazing as well as a permanent safety measure. Provide drawings as well as samples of the restraint clips.

3.5 Structural Requirements

Design all components of the system, including anchorages, fixings and adhesives, including structural adhesives, to carry all temporary and permanent design loads, individually and in combination without causing failure.

Failure shall be deemed to include cracking, bowing, distortion and looseness, lack of rigidity, dislodgement by wind forces, excessive deflection, visible movement at any joint, or defects which would cause damage to adjacent or applied work by others, and no permanent deforming of panels during installation or which will prevent panels to fit into place.

Materials and components shall be engineered to accommodate all loading conditions, and shall be of adequate size, strength and stiffness for the purpose intended.

Prepare and submit full structural analysis in the form of computations and shop drawings. Coordinate with the Main Contractor and Architect as required for additional information on base-structure conditions.

3.5.1 Design Loads

Refer to the **Façade Schedules** for minimum design loads.

Design loads are to be calculated in accordance with all relevant Codes and Standards and should cover the following:

- Dead loads
- Wind loads
- Live loads (both impact and maintenance loads)
- Earthquake loads



- BMU and roped access Loads

3.5.2 Structural Robustness

Refer ABCB Handbook – Structural Robustness 2020

The system during construction and use, with appropriate degrees of reliability is to be designed to sustain local damage, with the structural system as a whole remaining stable and not being damaged to an extent disproportionate to the original damage.

Structural robustness design may include notional removal of a load carrying member and a systematic risk assessment.

3.5.3 Temporary and Construction Loads

Design the system to include all handling and installation loads without causing over-stress, permanent deflection or warping.

No permanent deformation of panels, channel legs and the like during installation to enable panels to fit together will be approved.

3.5.3.1 VERTICAL LOADS TO HORIZONTAL FEATURES AND PARAPETS

All surfaces at 30 degrees or less to horizontal shall be designed for vertical point live loads in accordance with AS/NZS1170 .1. The point loading is to be supported by all façade components at all locations where human contact is possible.

3.5.4 Fatigue

Design the system to meet the fatigue requirements of the relevant design code e.g., AS/NZS 1664, AS 4100.

Wherever possible the following measures should be considered to minimise the possibility of fatigue failure:

- Avoid geometric stress concentrations or openings at high stress area/s.
- Incorporate smooth and gradual changes in geometry.
- Avoid fabrication techniques likely to cause fatigue e.g., welding.
- Connections should be designed to prevent the release or loosening of threaded joints due to mechanical vibrations or environmental factors e.g., Nyloc nuts, Loctite thread locker.

Systems that may be subject to wind-induced excitation from vortex shedding and buffeting, shall be assessed via full scale prototype testing in a wind tunnel.

3.6 Connection Requirements

3.6.1 Generally

Design and provide all connections, brackets and fixings by approved methods so that all loads will be transferred from the system to the base-structure in accordance with the design criteria in a manner which prevents excessive joint displacement, slippage or distortion.

There shall be no direct fixing or close contact between any part of the system and the base-structure or internal linings, except through approved bracket connections.

Design, schedule and indicate in detail on shop drawings, all types of dead load and restraint brackets and fixings. Indicate details of all related components and connections by other Trade Specifications.



3.6.2 Checking

Brackets shall be designed to provide three-dimensional adjustment and accurate location of the work and be rigidly fixed after the work is finally positioned.

3.6.3 Base-Structure Substrates

Provide contingency design and installation procedures for all typical substrate conditions and deficiencies including:

- Reinforcement clash.
- Excessive out-of-tolerance concrete.
- Honeycombing or other concrete defects.
- Clash with concrete joints, and other structural details.
- Incorrectly positioned, missed or incorrect embedments.

Check all base-structure shop drawings and concrete reinforcement drawings to verify bracket fixing locations. Indicate relationship between brackets and concrete reinforcement on shop drawings.

3.6.4 Anchors

Provide anchors and other methods of attachment of the system to the base-structure which:

- Fix the system in its correct position providing for base-structure and system movements.
- Are structurally adequate to carry the design loads, for the worst possible bracket positioning.
- Are compatible with the bracket assembly and provide three-way adjustment to accommodate fabrication and construction tolerances.

Excessive shimming will not be approved.

All post-installed and cast-in fastenings for use on concrete are to comply with AS 5216. Where a European Technical Assessment (ETA) is being provided to verify compliance, it must be relevant to the scope of the intended application.

3.6.5 Fixing Brackets

Fixing brackets are to be designed for the worst possible panel eccentricity, packing location, and uneven load sharing.

Design and install brackets so that all system loads are transmitted through brackets and prevent transfer of loads through panel joints to adjacent panels.

3.7 Movement Requirements

3.7.1 Generally

Design system to provide for deflections, displacements, and other movements within the system, and between the system and the base-structure (including fire stop and smoke flashing connections). Include movements caused by ambient temperature changes, wind loads, dead loads and live loads, and shrinkage.

The design shall include moveable joints to accommodate the full range of movement requirements including manufacturing tolerances, site tolerances, thermal movement, lateral movement, floor sag, beam sag, live load deflection and column settlement.

Allowance for movement shall be in addition to allowance for base structure tolerances.



Joints shall accommodate the worst possible combination of effects to prevent loads of any kind being transferred from the building into the system, excessive movements of any joints or failure of weather seals.

Indicate movement limits and required clearances on shop drawings.

All movement allowances shall be consistent and applied across all junctions and/or components for each expansion joint system or assembly.

3.7.2 Design Temperature Range

Refer **Façade Schedules**.

3.7.3 Base-Structure Movements

Design the system so that no structural or architectural element, joint or weather seal failure occurs due to base-structure movements. Refer Structural Engineer for Base-Structure Movements and **Façade Schedules**.

3.7.4 Settlement and Creep

Provide for differential movement between the façade system and the base-structure caused by all short- and long-term base-structure movements, settlement, and creep, including:

- Beam or slab deflections under designed dead and live loads.
- Column or frame shortening (elastic, creep, shrinkage, and the like)
- Differential settlement between any two columns or frames.

3.7.5 Thermal Expansion and Contraction of Base Structure

Provide for all thermal expansion and contraction of base structure.

3.7.6 Movement in the Base Structure Due to Wind Loads

Provide for differential movement between the façade system and the base-structure caused by lateral deflection under wind load of base-structure.

3.7.7 Movements within Façade System due to Thermal Expansion and Contraction

Provide for all thermal movement to take place freely in the plane of the façade system without causing harmful buckling, failure of joint seals, undue stress on fasteners and glass or other detrimental effects. No component or system shall be stressed or subject to induced loading as a result of thermal movement.

Full movement allowances including assembly and installation tolerances shall be incorporated into all junction/components at each expansion joint or assembly.

Provide for any additional movement within the façade system because of wind loads.

3.7.8 Structural Movement

Refer to Structural Engineer's documents for structural movement requirements.

Notwithstanding the above, it is the responsibility of the Façade Contractor to refer to the appropriate specifications of Cast-In-Place Concrete, Concrete Formwork, and Structural Steel; and to confirm with the project Structural Engineer for the full description of structural movements.



3.8 Environmental Requirements

3.8.1 Generally

Design and warrant the system to be watertight and weatherproof.

Provide for all weather conditions. Submit complete details on shop drawings including all assumptions.

All sealing, gaskets and flashings, including concealed and inaccessible items, shall prevent penetration of water or moisture to any portion of the interior of the finished building.

3.8.2 Control Devices

Water which enters the system, including leakage and condensation shall be drained or discharged to the exterior face of the system. Water shall not be trapped or retained in the system. Risk of algal or fungal growth shall be minimised.

Sealants and gaskets which are inaccessible in the installed system shall remain effective for the service life of the system.

3.8.3 Condensation

Profiles shall provide for positive removal of condensation. Provide for natural weeping of all condensation to the outside at head sections or sill sections. Condensation on vertical surfaces shall weep out at sill line.

The Façade Contractor must manage water vapour and condensation for frequency of occurrence and severity with regards to building classification, occupancy, and HVAC provision.

3.8.4 Self-Generated Noise

Design and install the system generally and all component parts to provide for noiseless movement caused by thermal expansion and when subject to dynamic load caused by wind, and cleaning operations. The system shall not generate noise due to whistling, creaking, drumminess or rattle.

3.8.5 External Reflectivity

All materials shall comply with external reflectivity limits where they have been stated in Planning Conditions or the requirements of a Reflectivity report, if one provided for the project.

3.8.6 Thermal Performance

Computations verifying compliance of the proposed façade system with the mechanical performance requirements specified by the Mechanical Engineer and noted in the **Façade Schedules** are to be submitted by the Façade Contractor. This is to include the following:

- System U Value Calculations
- Verification of Glass Performance VLT and SHGC

3.8.7 Lightning Protection

Refer to the requirements specified by the Electrical Engineer for lightning protection.



3.9 Fire Performance

All elements of the façade system are to be deemed non-combustible in accordance with the NCC and where required, verified through testing in accordance with both AS1530.1 and AS 5113. Testing is to be conducted by an ILAC, NATA or HOKLAS accredited laboratory. Test results are to be submitted as a technical submission for approval by the Design Team, Fire Engineer and Building Surveyor.

Samples and technical data of all materials in the façade system are to be submitted to the Building Surveyor and Fire Engineer for their review.

3.10 Maintenance Requirements

3.10.1 Generally

Design the system to ensure that the service life and all product warranties can be properly fulfilled as a result of reasonable and adequate cleaning and maintenance procedures.

Submit details including computations for additional loads and deflections due to cleaning and maintenance operations.

Co-ordinate design of the façade system with safe access requirements. Allow for all maintenance loads in façade system design.

3.10.2 Replacement Methods

Design and install the system to allow for the future replacement of any individual panel. Façade Contractor must submit details and method statements for replacing individual damaged items as part of shop drawing/design submission.

3.10.3 Damage

Horizontal or near horizontal surfaces which may carry human live loads from maintenance personnel, shall be capable of supporting such loads without permanent distortion or other damage.

3.10.4 Self-Cleaning

The design of all components in this contract shall be such that their configuration shall facilitate the self-cleaning of the various systems. Inclined drainage surfaces and flashings, properly designed drip grooves and self-cleaning joint design are examples of this requirement.

3.10.5 Removable/Demountable Panels

Design and install the removable/demountable panels for down pipes and asset replacement and others as required.

3.10.6 Infestation

All materials used within the façade system shall not be damaged or infested by vermin, insects and plant-life such as fungi etc.



3.11 Visual Requirements

3.11.1 Profiles

Design all components so that architectural styles, sizes, profiles and dimensions are as close as possible to those indicated on the Contract Documents. Indicate any proposed variations on the shop drawings. Do not commence production of extruded aluminium sections until all profiles have been approved in writing.

3.11.2 Visual Mock-Up (VMU) / Prototype

A visual prototype is to be provided as specified in the **Façade Schedules**. The visual prototype shall be full size and shall include the actual glass type, frame material and finish.

The Visual Mock-up is to be viewed:

- Over a range of viewing angles which may require the VMU to be lifted on a forklift so it can be viewed obliquely.
- Under Australian blue-sky conditions and overcast/fading light conditions.
- A backing sheet or enclosing room is to be fixed to the back of the panels for background uniformity in a fashion representative of the project conditions.
- Orientation to reflect project site conditions.

Should the Visual Mock-Up be located outside of Australia, the Façade Contractor shall include provision for:

- The Client, Superintendent and the Architect to be present to review and approve it and,
- The glass to be sent to Australia for review and approval.

3.11.2.1 VISUAL MOCK-UP (VMU) – GLASS SAMPLES

The glass samples submitted for approval must be produced by the same coating production line as that which is supplying glass for the project.

The glass samples are to be viewed:

- Supported and orientated on framework to reflect project site conditions and,
- A removeable backing sheet is to be fixed to the back of the panels (or framework supporting panels) for background uniformity in a fashion representative of the project conditions.

Unless otherwise approved, glass will only be approved under Australian sky conditions.

3.12 Tolerances

The Façade Contractor shall ensure that all tolerances are considered in their design. These tolerances include but are not limited to the following:

- Building tolerances
- Fabrication tolerances
- Installation tolerances

Refer to the **Façade Schedules** for proposed allowable tolerances to be agreed with the Main Contractor.

3.12.1 Building Tolerances

Provision shall be made in the design of the façade system for the erection tolerances of building structures. The Façade Contractor shall be responsible for agreeing to all tolerances with the Main Contractor.



Notwithstanding the above, it is the responsibility of the Façade Contractor to refer to the appropriate specifications of Cast-In-Place Concrete, Concrete Formwork, and Structural Steel for the full description of building tolerances, and to coordinate tolerances to be expected with the Main Contractor prior to Contract Award.

The concrete shall be designed to accommodate the primary structure construction tolerances allowed to the Façade Contractor. Unless otherwise noted, the maximum construction tolerances shall be as specified in the **Façade Schedules**.

3.12.2 Metal Fabrication Tolerances

Metal fabrication tolerances shall be such that there will be no visible oil canning. Cut edges, drilled holes, riveted joints and flat sheets shall be clean, neat, and free from burrs and indentations. Remove sharp edges without excessive radiusing.

3.12.3 Installation Tolerances

The Façade Contractor is required to site check the cumulative effect of all building tolerances and adjust installation procedures as required. All installed façade system units and components shall be subject to measurement of tolerances during fabrications, assembly, storage, installation and post-installation.

3.13 Safety in Design

The Façade Contractor is required to submit a Safety in Design (SID) review with submission of shop drawings and engineering calculations, in accordance with OHS legislation (Victoria and WA) or the model WHS laws (all other states and territories) where applicable.

In developing the design, the Façade Contractor is required to take account of the risk assessment carried out at tender stage.

Refer **Façade Schedules** for this record, which is based on information provided at the time of specification issue.



4.0 Materials

All materials are required to be certified prior to commencement of fabrication. The Façade Contractor is required, when requested, to provide Certification from the manufacturer of each material stating that the material provided is as per the specified material and is in accordance with the relevant standards.

Materials where certification may be requested for include:

- Glazing
- Metal cladding
- Metal framing
- Insulation and fire stop materials
- Bolts, screws and other fixings
- Door and window hardware – where applicable
- Sealants, tapes, setting blocks and gaskets

4.1 Glass

Refer to the Drawings for glass types and locations and to the **Façade Schedules** for glass types.

All glazing shall be engineered to comply with all relevant Codes and Standards and should consider:

- Structural adequacy
- Serviceability
- Section J Compliance
- Thermal stress adequacy
- Acoustic requirements

Computations verifying the above are, unless otherwise advised, required to be submitted for review.

The Façade Contractor is also required to submit documentation verifying that the glass supplier has obtained ISO 9001 accreditation.

4.1.1 Thermal Properties

In complying with the National Construction Code, all glass products must be tested under NFRC 100 conditions and the technical performance data be verified by either:

- Being registered on the International Glass Database (IGDB – accessible on WINDOW 6.3 or later) – reference number to be provided; or
- Through being tested by an accredited independent testing authority.

As per the above, should the proposed glass not appear on the IGDB, the Façade Contractor is required to allow for the cost and submit test data from an appropriate test facility (**Refer Section 2.10.2.1 – Testing Facility**) indicating the following properties:

- Visible light transmittance (VLT)
- Visible light reflectance (VLR) (internal, external)
- Shading coefficient (SC) (NFRC 100, ASHRAE summer)
- Solar Heat Gain Coefficient (SHGC) (NFRC 100, ASHRAE summer)
- U-value (NFRC 100, ASHRAE winter)
- Solar transmission
- Solar absorption



- Solar reflectivity

4.1.2 Quality Requirements

All flat glass shall be approved float glass in accordance with ASTM C1036, Type "Q3".

All glass shall be free from cracks, scratches, bubbles, blisters, all inclusions of deleterious matter including nickel sulphide and other defects which detract from appearance or interfere with durability and performance. Defects are required to comply with AS/NZS 4667.

Production and test records to ASTM 1036, Table 4, proportionally adjusted to proposed project glass thickness are required to be submitted. The reports should indicate the test procedures, acceptance criteria and pass and/or fail rate.

Include manufacturer's data related to nickel sulphide control.

4.1.3 Low Iron glass

Low Iron or ultra-clear glass shall be made from low-iron silica sand typically with a ferric oxide content below 0.01%.

The colour rendering index (CRI) $R_a > 96\%$ measured as per BS EN 410: 2011 - Glass in building. Determination of luminous and solar characteristics of glazing.

Visible Light Transmittance benchmarked on a 6 mm thick glass $> 91\%$

Suitable Low-Iron products that may be approved include: Starphire (Vitro PPG), Optiwight (Pilkington), Diamant (Saint-Gobain), Ultra-clear (Gardians).

Sample is subject to approval by the Architect.

4.1.4 Annealed Glass

All annealed glass shall have a minimum thickness of 6 mm in monolithic and 6.38mm in laminated assemblies.

All glass shall be in accordance with ASTM C1036 Table 2 and have clean, wheel cut edges with minimum feather, free from vents, notches or shells.

Any undulation distortion of the installed work shall be horizontal unless otherwise approved in writing. The maximum undulations shall not exceed 0.08 mm in any 300 mm length of glass.

The use of annealed glass will only be accepted if it can be demonstrated that it satisfies thermal breakage risk for all conditions including all weather conditions, internal operating conditions, impacts from blinds and other shading devices and operating conditions (including the placement of objects adjacent to the glazing).

Thermal stress analysis to be undertaken in accordance with the Pilkington Method or ASTM E2431.

4.1.5 Heat Strengthened and Toughened Glass

The base material shall be an approved selected quality annealed glass manufactured using the "roller hearth", or an equivalent approved process in accordance with the relevant Standards.

Heat-strengthened properties of glass will be verified via non-destructive testing to confirm the surface compressive stress is between 30 MPa and 52 MPa.

Toughened properties of glass will be verified via non-destructive testing to confirm the surface compressive stress is greater than 69 MPa.



4.1.5.1 TOUGHENED GLASS

All toughened glass shall be 100% heat soak tested in accordance with EN 14179 Part 1. Heat soaking method, temperature calibration method and duration to be submitted for approval prior to commencement.

Do not use monolithic toughened glass:

- Unless specifically noted in the contract documents or approved by the Client or Client's nominated representative or,
- Where in the event of breakage, glass could fall from a height of more than 3m.

Where toughened glass has been selected by the Façade Contractor, the Client shall be formally notified of the residual risk of failure, following a rigorous analysis of alternative glass configurations by the Façade Contractor and Main Contractor. Where feasible, toughened glass should be designed out due to the inherent risks and costs associated with spontaneous glass breakage and glass replacement.

4.1.5.2 DISTORTION

Any roller wave distortion to the installed work shall be horizontal unless otherwise approved in writing. The maximum roller wave shall not exceed 0.15 mm in any 300 mm length of glass.

Bow distortion (also referred to as 'warp' distortion) shall have the same meaning. Overall bow across the shorter glass dimension ('the gauge length') shall not exceed 0.00342 times gauge length for gauge length less than 900mm, and 0.00258 times gauge length for gauge length greater than 900mm.

Edge curl shall be limited to less than 0.2 mm.

4.1.5.3 EDGE QUALITY

All concealed edges of tempered glass panels shall be bevelled edges. All exposed glass edges are to be mitred and polished.

Do not cut, work, or permanently mark after toughening. Use installation methods which prevent the glass making direct contact with metals or other non resilient materials.

4.1.6 Laminated Glass

Laminated glass shall be in accordance with ASTM C1172, ISO 12543 and comply with the requirements of AS/NZS 2208.

Suitable interlayers that may be acceptable are Kuraray SentryGlas Plus or Trosifol PVB.

Details including test results for adhesion, bake test, durability and compatibility with proposed sealants are required to be submitted.

Use glazing materials which do not cause deterioration or discolouration of the interlayer.

Glass panels shall be vacuum bagged for autoclaving. The use of glass clamps is not permitted.

All secondary manufacturing of glass shall be carried out by a factory certified under ISO 9001. Submit production and test records. Indicate test procedures, acceptance criteria and pass and/or fail rate. Include records for interlayer. Confirm that the interlayer manufacturer's requirements have been fully complied with e.g., relative humidity, temperature, pressure and temperature, cooling rate, etc.

4.1.6.1 SOLAR CONTROL INTERLAYER

A suitable solar control interlayer that may be acceptable is Eastman XIR solar control film.



XIR film shall be used only for flat laminated glass.

4.1.6.2 ULTRA-CLEAR PVB INTERLAYER

When an ultra-clear PVB interlayer is specified, the PVB interlayer must demonstrate a yellowness index of 0.8 or below for a 1.52 mm interlayer thickness, when determined in accordance with ASTM E313.

A suitable ultra-clear PVB interlayer that may be acceptable is Ultraclear Trosifol (Kuraray).

4.1.6.3 IONO-PLAST INTERLAYER

Suitable interlayers that may be acceptable are Kuraray SentryGlas Plus

Thickness of interlayer to be approved by supplier and as required by glass heat treatment.

4.1.7 Insulating Glass Units

Provide insulated glass units (IGUs) (also known as double glazed units) of approved types in accordance with the relevant Standards. Submit complete details, including all construction details, primary and secondary seals, gas filling, corner treatment, nature of spacers, evacuation and assembly process.

4.1.7.1 SEAL PERFORMANCE

IGUs shall incorporate approved polyisobutylene primary (vapour) seals continuously bonded to glass, and two-part silicone secondary (structural) seals. Primary seal shall be not less than 3 mm deep. Secondary seal shall completely cover spacer with no gaps or voids, continuously bonded to glass.

Spacer bar shall be an approved proprietary aluminium type, desiccant filled, anodised in an approved colour compatible with seals.

4.1.7.2 STRUCTURAL EDGE DISTANCES

Submit computations for distance between the outer edge of the glass and the innermost edge of the aluminium spacer indicating minimum bite of secondary (structural) seal and spacer bar depth. Include sealant manufacturer's recommendations and load tables.

4.1.7.3 EDGE QUALITY

External heat strengthened lites of IGUs shall have clean cut edges. Internal tempered lites of IGUs shall have bevelled edges.

There shall be no exposed edges to IGUs in finished installation.

Sight lines shall be clean and straight to match approved sample.

4.1.7.4 QUALITY RECORDS

All secondary manufacturing of glass shall be carried out by a factory certified under ISO 9001. Submit production and test records for IGUs. Indicate test procedures, acceptance criteria and pass/fail rate to relevant Standards.

4.1.7.5 TESTING

Submit a report from a testing laboratory that the IGU type has been tested and passed the requirements of one of the three suites of standards (ASTM, ISO or EN) nominated in AS4666.



4.1.7.6 WARRANTY

All IGUs complete shall be warranted for the warranty period. All IGUs shall be free from evidence of manufacturing defects and shall be free from:

- Intrusion of moisture or dirt.
- Internal condensation at temperatures above -10°C.
- Other visual evidence of seal failure or performance failure.

4.1.7.7 DESIGN

All IGUs shall be designed to resist the combined effects of changes in temperature, altitude, atmospheric pressure together with the other specified loads.

4.1.8 Low-E Coated Glass

Low emissivity (Low-E) coatings shall be neutral in transmitted and reflected colour and otherwise exhibit the visual and performance characteristics of the products specified.

The coating shall be approved by the Architect.

Visual Quality Control acceptance criteria of the Low-E coating shall be consistent with industry guidelines, subject to approval by the Architect and the Principal.

- Pinholes with diameters in excess of 1.5 mm are not acceptable.
- Clusters of pinholes are not acceptable.
- Scratches no longer than 75 mm in length are acceptable provided that they occur within 75 mm of an edge.
- Colour evenness within each glass pane shall not exceed $\Delta E^*_{ab} \leq 2.5$, and the colour difference between glass pieces in the supply for the whole project shall not exceed $\Delta E^*_{ab} \leq 3.0$.

Edge deletion of Low-E coatings shall be provided at all insulating glass and structural glazing unless manufacturer submits test data acceptable to the Architect indicating that edge deletion is not required.

The Façade Contractor shall ensure that the coating conforms to the codes and regulations. Any fees incurred from an application of waiver and variance shall be the responsibility of the Façade Contractor.

The manufacturer shall warrant that the Low-E coating shall not peel, crack, fade or deteriorate for the warranty period. The manufacturer shall warrant that any glass that is not edge deleted will not develop loss of adhesion with insulating glass or structural glazing sealants for the warranty period.

4.1.9 Production Quality Control for Quenching Heat Treated Glass

All heat treated glass is to be produced in a manner that prevents uneven quench patterns in the glass, so as to prevent the blotchy appearance or colour change when viewed with the naked eye from any angle and distance under a typical range of daylight conditions.

4.1.9.1 GLASS MANUFACTURE

Prior to glass manufacture, the Façade Contractor is to provide QA documents from the glass factory to show the glass will be checked for quench patterns at the commencement of production for every glass configuration, and then at minimal intervals as outlined below:

- Order size less than 100 panels - 2 hour intervals
- Order size 101 to 500 panels - 4 hour intervals



- Order size over 500 panels – daily

The Superintendent may request that the glass QA inspection records during the manufacture process are available for review and approval.

Refer also 3.11.2 Visual Mock Up for glass viewing and approval of visual mock up.

4.1.9.2 GLASS QUENCHING – PASS/FAIL

Glass panels will be assessed with a polarising light filter to a digital camera which will be used to identify quench pattern bands (only). These will then be checked by naked eye, over a range of light conditions and angles to replicate exposure conditions on the building.

Glass will be rejected on the basis of there being uneven quench patterns when blotches are visible by naked eye from any viewing angle. If some degree of unevenness is unavoidable, the Client / Superintendent/Architect are to be provided with a sample as a benchmark for their consideration. If this is accepted it shall be used as a control sample for production.

If uneven quench patterns are observed, production through the oven used to heat treat the glass is to be ceased, and the heat treatment oven is to be checked/ serviced by the furnace manufacturer. Where thermal imaging cameras are fitted, glass thermal images are to be compared with quench patterns to determine whether there are areas of over-heating to the glass.

Following servicing of the oven, test panels are to be manufactured and tested prior to proceeding with production.

4.1.10 Ceramic Enamelled (Fritted) Heat Treated Glass

Ceramic ink can be applied to area of glass using silk screen or digital printing techniques. The printed surface shall face the inside of the building.

Ink shall be fused to the glass surface during the heat treatment process and provide crack resistance up to 250°C. Ink shall be permanent, durable and resistance to any mechanical damage or abrasion in the normal use and maintenance procedure during expected design life of the façade. Ink shall be resistant to weathering, fading, or discolouration due to climate condition, UV and atmospheric pollutants.

Ceramic ink shall be free from lead, lithium, cadmium or any other substances that are considered hazardous when released into the environment during manufacture, in use or during demolition and re-cycling.

Ink colour shall be matched to the RAL Colour System or other approved colour range. Submit samples and confirm colour with Architect prior to manufacture. Ink coverage shall be to within 3mm of the edge of the glass.

Defects, including scratches and pinholes shall not be visible when viewed at a distance of 3m.

Cutting of panels, edge distance for colour coverage and ceramic pattern orientation shall be closely co-ordinated. Confirm pattern / layout with Architect prior to the start of mass production.

Tolerance of silkscreen	+/-2 mm in all direction
Minimum gap size between dots	3 mm
Minimum dot size	3 mm

100% fritted glass shall not be permitted for back-lit applications e.g. lightbox, unless otherwise approved.



Where ceramic enamelled (fritted) glass is in direct contact with laminating interlayer (e.g. pvb, sgp), the Façade Contractor is required to obtain in writing, confirmation from glass supplier that structural adequacy of laminated glass, and integrity of frit and interlayer is not compromised.

Where ceramic enamel (frit) is applied to low-E coated glass, the Façade Contractor is required to obtain in writing, confirmation from glass supplier that integrity of coating and frit is not compromised.

4.1.11 Warping of Panels

Where design requires warped panels, it is assumed that glazed units will be fabricated flat and then warped (along with the façade frame) on site during installation or alternatively, the framing may be fabricated in a pre-formed shape to which the glass would then be combined.

Warping of panels shall not cause any visible distortion to either framing or cladding elements.

4.1.11.1 CUTTING

Glass for bending shall be wheel cut on a digitally controlled cutting table to dimensions including any allowance for bending.

4.1.11.2 REQUIRED SUBMISSIONS

The Façade Contractor shall provide the following submissions, other than the requirements for glass generally, referencing the degree of curved glass present in the Works:

- Stress analysis of individual glass panels
- Stress analysis within the hermetic seal of the insulated glass units
- Spacer design and commentary on how it responds to the geometry of units in the fabrication process
- Stress analysis of silicone seal between glass lites
- Stress analysis of the interlayer material
- Letter of warranty from the supplier of the insulated glass units (and associated components) specifically noting the curving of panels
- Letter of warranty from the structural sealant manufacturer specifically noting the curving of the glass
- Letter of warranty from the interlayer supplier and glass supplier for curved glazing.

4.2 Elastomeric Extrusions, Gaskets, Seals, Setting Blocks & Accessories

Provide approved non-cellular elastomeric extruded profiles, including gaskets, and seals required for a complete installation, in accordance with relevant standards. Submit details, including compatibility with adjacent components and sealants. Indicate sizes and locations on shop drawings.

Extruded profiles shall be smooth, of uniform dimensions, correctly selected for the conditions of use, and free from components likely to bleed, stain or detrimentally affect performance of adjoining materials in the façade system. All products shall be of ultra-violet resistant grade.

Exposed sealants will not be approved, except where indicated on approved shop drawings.

Gaskets and dry weather seals shall be Silicone, EPDM, Neoprene or approved equivalent.

PVC gaskets may not be used without prior approval.

Setting blocks for glazing shall be dense heat cured silicone rubber with a Shore A hardness of 80-90 durometer. Side blocks shall be dense heat cured silicone rubber with a Shore A hardness of 60-70 durometer.



4.2.1 Quality

Submit production and test records for all products in accordance with relevant Standards. Indicate test procedures, acceptance criteria and pass and/or fail rate.

4.2.2 Warranty

All products shall be warranted for the Warranty Period – refer also to the **Façade Schedules**.

The warranty shall include specific provisions and terms for failure of, or due to, the following:

- Cracking due to ozone and ultra-violet attack or any other reason.
- Sustaining growth of mildew, mould, fungi and the like.
- Excessive discolouration due to exposure to sunlight and weathering.
- Excessive deformation under load.
- Excessive ageing, detrimental increase in hardness or change in tensile strength and elongation more than allowable under ASTM C864.

4.2.3 Co-Extrusions

Where indicated on the Drawings or required, provide approved co-extruded profiles of polypropylene and polypropylene compatible products.

Products which may be approved include those manufactured from EPDM or equivalent.

4.2.4 Hard Profiles

Where indicated on the Drawings or required, provide dense profiles including flashings, wiper seals and the like, complying with the relevant Standards, as follows:

- Shore A durometer hardness: 75 ± 5 for solid profiles and 60 ± 5 for hollow profiles when tested in accordance with ASTM D2240.
- Compression set 100% (168 hrs): Not greater than 40% when tested in accordance with ASTM D395.

4.2.5 Soft Profiles

Where indicated on the Drawings or required, provide soft profiles including bulb seals, sponge seals and the like, complying with the relevant Standards, as follows:

- Shore A durometer hardness: 40 ± 5 when tested in accordance with ASTM D2240.
- Compression set (168 hrs): $30\% \pm 5\%$ when tested in accordance with ASTM D395.

4.3 Structural Silicone

Structural silicone to be applied under controlled factory conditions.

Site applied structural sealant, except as required for glass replacement, will not be permitted unless noted otherwise.

Manufacturers of structural silicone that may be acceptable include: Dow Corning (Dowsil), General Electric, Tremco, Sika Chemicals.

All products proposed shall be known and proven products.



4.3.1 Structural Silicone Glazing of Glass Butt Joints

Structural silicone to be compatible with interlayers and test reports and letter from silicone supplier provided for verification.

Site applied; well-tooled approved silicone applied by fully trained personnel. Installation with proper surface preparation, clean and dry. Curing temperature and dry/moisture-controlled conditions, clean of dirt. Silicone fir tree gasket to be pressed between the front and rear 2:1 width to depth ratio.

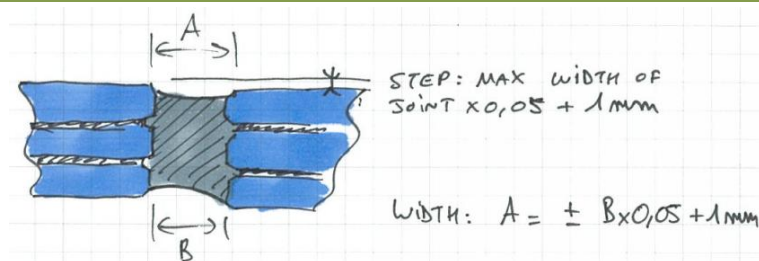
First, fully tape the side of the glasses to protect and do a neat straight finish to the silicone joints (no wavy edges). Fill the gap of the interlayer first, using same structural silicone and pushing the silicone in the gap with a flat trowel. Straight after before it cures, fill with silicone the gap, tool it flush. Ensure no bubble, full contact with the silicone. If three ply laminate or more, the silicone must be tooled concave and then after curing, finish flush with a second silicone top-layer, both faces. Protect the works from dust until the top-layer is cured.

Butt joint above 4-meter length: Shear strength/stiffness calculation taking into account the aspect ratio (Dow Corning GPD method) to panel coupling (locking differential deflection).

Silicone thickness above 25 mm: Curing efficiency through thickness and curing procedure to be detailed from the supplier.

Compatibility testing of the structural silicone with glass interlayers.

Tolerances on joints:



4.4 Weatherproofing Silicone

Manufacturers of low modulus silicone sealant that may be acceptable include: Dow Corning (Dowsil), General Electric, Rhone Poulenc, Tremco, Wacker, Sika Chemicals.

Weatherproofing sealant to be the same brand as structural silicone sealant.

Weatherproofing sealant shall be non-staining and compatible with all substrates, including interlayer of laminated glass.

Colour:

- Weather seals: Black or as specified
- Air seals: Black or as specified
- Butt joints: Clear or as specified



4.5 Thermal Barriers used in Metal Frames

The thermal barrier shall be composed of non-metallic materials (e.g. polyamide, polyurethane, improved synthetic material) conforming with the requirements of BS EN 14024 or AAMA TIR-A8. Provide details of thermal barrier composition for review.

Suitability of the thermal barrier material shall be assessed and demonstrated through testing, depending on the mechanical design of the system, which may include (but is not limited to):

- Performance after immersion in water
- Performance after exposure to humidity
- Testing for tensile cracks
- Testing for brittleness
- Transverse tensile strength
- Shear strength and elasticity constant
- Ageing

Compatibility of sealants with the thermal barrier shall be demonstrated through testing (if relevant) or through a letter from the sealant manufacturer, if compatibility data available.

4.5.1 Mechanically Locked Preformed Thermal Barriers

The thermal coefficient of expansion of the thermal barrier material shall be matched to that of aluminium, to avoid dry shrinkage.

4.5.2 Poured and Debridged Thermal Barriers

For poured and debridged thermal barriers, adhesion adequacy between the surface treatment on the interior cavity surface of the metal profiles to the polyurethane shall be demonstrated through thermal cycling testing to AAMA 505 (or approved equivalent).

Refinishing processes that occur after filling and debridging shall not be permitted unless sufficient evidence is provided to demonstrate that system performance is not adversely affected, and agreement is provided from the thermal barrier supplier.

4.6 Steel

Refer to the Steel Specification by the Structural Engineer.

All steel shall comply with AS 4100 and all other relevant codes, standards, and regulatory requirements.

4.6.1 Steel Fabrication

The work shall consist of the design, supply, fabrication, surface treatment, storage, delivery and erection of all the steelwork shown on the contract drawings. This also includes the supply and installation of all cast-in items used to support the steelwork, the grouting of base plates, the provision of cleats and drilling of holes for the attachment of the façade system, and repairs to surfaces damaged during construction.

The work shall be carried out in strict accordance with the Registered Structural Engineer's drawings, this specification, all other drawings and specifications included in this contract, and any authorised written instruction issued by the Architect during the course of this project.

Where any conflicts are found in the contract documents by the Façade Contractor they shall be immediately brought to the attention of Main Contractor, Superintendent and Architect. The Façade Contractor shall await



a written instruction from Main Contractor, Superintendent and Architect before proceeding with the affected area of work.

Steel fabrication tolerances shall be as per the **Façade Schedules** unless otherwise approved.

4.6.2 Friction-Type Bolted Connections

Prepare contact surfaces of friction-type bolted joints by wire brushing or light grit blasting to achieve the required slip factor.

4.6.3 Protective Coatings to Above Ground Steelwork

All steelwork is to be provided with a protective coating system to meet the minimum design life as stated in the Specification or **Façade Schedules** unless otherwise advised. The coatings are to be shop applied and provided over the full surface area of steelwork. Exposed edges and site weld areas are to be coated with an approved coating system for onsite application. Details are to be submitted for approval. Refer also to the **Façade Schedules**.

4.6.4 Paint Coatings to Above Ground Steelwork

All painting works shall be applied in accordance with this specification and the requirements of AS/NZS 2312.1.

4.6.4.1 SUBMISSIONS

Coatings shall be used in accordance with the manufacturer's recommendations and materials safety data sheets, unless otherwise specified in this Specification.

The Façade Contractor shall submit for review by the Client or Client's Nominated Representative not less than twenty one (21) days prior to the commencement of the coating works, details of the steel coating operations including information on the proposed coatings, surface treatments, substrate preparation, method of application, equipment and operators, demonstrating compliance with the requirements of this specification.

Material details shall include information on the properties of the proposed products, documented evidence of previous performance and relevant test results which shall not be more than twenty-four (24) months old.

Coating works shall not take place until the Façade Contractor's proposed materials and procedures and surface preparation have been reviewed and accepted by the Client or Client's nominated representative.

4.6.4.2 COATING SYSTEM DETAILS

The Façade Contractor is to engage a suitable professional to select the coating system to be used based on the following:

- Design life of the coated article in accordance with ABCB Handbook: Durability in Buildings including Plumbing Installations or NZBC Acceptable Solutions and Verification Methods - B2 Durability or another relevant Standard.
- Atmospheric Corrosivity category as defined in AS/NZS 2312.1
- Micro environment conditions on site e.g. condensation risk, cyclic changes, ground contact
- Surface preparation to BS EN ISO 8501-1
- Assessment of life to first maintenance in accordance with AS/NZS 2312.1
- FRL requirement (if relevant) in accordance with the Fire Engineer's Specification
- Requirements as per **Façade Schedules**
- Structural Engineer's Specification (if relevant)



- Architect's External Finishes Schedule (if relevant)

The life to first maintenance period should match or exceed the design life of the coated article wherever feasible.

Where there is a contradiction in requirements, the more onerous condition applies, unless approved/instructed otherwise by the Client or Client's nominated representative.

Extract from ABCB Handbook: Durability in Buildings including Plumbing Installations (2015)

3.4 Minimum Design Life

The minimum design life for a building or plumbing installation and the components of their subsystems should be as shown in Table 3-1.

Table 3-1 Design life of building and plumbing installations and their components

Building Design Life Category	Building Design Life (years)	Design life for components or sub systems readily accessible and economical to replace or repair (years)	Design life for components or sub systems with moderate ease of access but difficult or costly to replace or repair (years)	Design life for components or sub systems not accessible or not economical to replace or repair (years)
Short	1 < dl < 15	5 or dl (if dl < 5)	dl	dl
Normal	50	5	15	50
Long	100 or more	10	25	100

Note: Design Life (dl) in years

The design life of buildings should be taken as 'Normal' for all building importance categories unless otherwise specified.

The coating system shall be a pigmented coating to an agreed colour. All components in the system (e.g. primer, undercoat and finish coat) shall be supplied by the same manufacturer. Colour and finish to be coordinated with the Architect's External Finishes Schedule (if relevant). Details are to be submitted for the Architect's approval.

Some colour finishes may require multiple coats to achieve adequate opacity.

A stripe coating shall be applied to ensure that an adequate dry film thickness is achieved on edges, corners, welds, pitted surfaces, flanges, nuts, bolts and other irregular surfaces. To overcome problems of pull-back and draw-off from such surfaces, an additional coat of primer or intermediate coat (referred to as a stripe coat) must be applied. The stripe coat is to be brush applied for welds and corroded / pitted surfaces.

The coating system shall be capable of withstanding cleaning with hot water, in the range of 40°C to 50°C, detergent and scrubbing action without loss of adhesion, softening or changing in colour or gloss.

The Contractor shall provide evidence as to the compatibility of the proposed coating system to the nominated substrate. Such evidence shall include but not be limited to test certificates from a NATA accredited testing laboratory and history of previous performance.

Where a primer or undercoat is required as part of the coating system, the primer or undercoat shall be a different colour to that of the final nominated colour and shall be in accordance with AS 2700.



The coating pigments shall be colourfast, and not subject to fading or discolouration.

4.6.4.3 PAINT HANDLING AND STORAGE

In accordance with the requirements of AS/NZS 2312.1.

4.6.4.4 SUBSTRATE SURFACE PREPARATION

4.6.4.5 GENERAL

The surface preparation shall be in accordance with the manufacturer's recommendations for the coating system to be applied and as specified in this Specification.

All surfaces must be cleaned free of oil and grease before processing using a neutral/alkaline degreasing process in accordance with AS 1627.1.

4.6.4.6 ABRASIVE BLAST CLEANING

In accordance with the requirements of AS/NZS 2312.1.

All existing sharp edges and corners are to be rounded and laminations, cracks, weld spatter, slag deposits and flux residues removed by the Façade Contractor prior to blast cleaning.

4.6.4.7 APPLICATION OF COATINGS

In accordance with the requirements of AS/NZS 2312.1.

Painting must be carried out on prepared surfaces before any discolouration occurs, i.e. the degree of flash rusting present shall be nil. Unless otherwise agreed the time lapse between cleaning and the commencement of coating must be within the same day or shift. The Contractor shall be responsible for taking appropriate precautions to protect cleaned surfaces from recontamination.

Coating systems shall not be applied under any of the following conditions:

- windy conditions where over spray and/or spatter may be generated;
- when wind-borne debris is likely to contaminate the uncured surface of the freshly applied coating;
- when the ambient temperature exceeds 35°C or is below 10°C unless otherwise expressed in writing by the manufacturer;
- when the relative humidity exceed 85% or where it may be expected to exceed 85% during the subsequent 12 hour curing period;
- when rain spatter or run-off contaminating the surface and adversely affecting the adhesion to the substrate, may occur;
- when the substrate surface is wet or damp;
- the surface temperature of the substrate is less than 3°C above the dew point or exceeds 40°C.

The minimum and maximum allowable drying times between coats, the minimum number of coats and the minimum DFT for each coat and for the whole system shall be in accordance with the paint manufacturer's requirements.

The wet film thickness shall be checked in accordance with ASTM D4414 84:2007. The dry film thickness shall be checked in accordance with AS 1580.108.2.

The Contractor shall protect already painted or galvanised surfaces, services, joints, painted signs and nameplates during the surface preparation process and during coating application processes.



4.6.4.8 DRYING AND CURING

The Contractor shall adhere to the manufacturer's instructions regarding drying and curing requirements, and overcoating time intervals, for the prevailing weather conditions.

4.6.4.9 CLEAN UP OF WORKS AND PROPERTY

The Contractor shall remove all coating drips and droppings, smudges and over spray from all surfaces, including surfaces not being treated. The Contractor shall remove from the site all debris and other rubbish accumulated during the work. The Contractor shall dispose of such wastes to the satisfaction of the Principal and by a means which conforms to Section 4.6.4.10 Waste Disposal.

The Contractor shall ensure that the coated works are protected from adverse conditions, dust and debris during the curing period of the coating system.

No spray painting shall be carried out within ten (10) metres of buildings, footpaths, roadways, pedestrians or vehicles without protective measures or methods being used which shall be submitted to the Principal for approval, a minimum of two working days in advance of the proposed works.

4.6.4.10 WASTE DISPOSAL

The Contractor shall adhere to Environment Protection Authority and other local, state and federal government requirements with respect to how waste generated during surface preparation, coating application, and clean-up will be collected, segregated, handled, controlled and disposed of.

Waste materials including liquid wastes shall be deposited in suitable containers and disposed of at sites to be located by the Contractor that are acceptable to the Environment Protection Authority and other relevant authorities.

Liquid or other waste material shall not be disposed of in creeks, waterways or the stormwater drainage systems.

4.6.4.11 INSPECTION AND TESTING REQUIREMENTS

The Contractor shall undertake all inspection and testing as specified in this clause. The Contractor shall maintain all required documentation and results as specified in this Specification.

The work shall be inspected by the Contractor at each stage of the coating operation as a minimum, i.e. after surface preparations, prior to each coat and after final top coat application and any touch-up coats that may be required.

For the purpose of surveillance and audits, the Contractor shall give the Principal five (5) days written notice of its intention to coat the steel and shall provide adequate access to enable surveillance of the Works by the Principal.

4.6.4.12 LEVEL OF INSPECTION

The level of inspection required shall be Level 1 in accordance with AS/NZS 2312.1:2014 Section 9.3 where the Contractor carries out the inspection requirements as part of their quality assurance plan. No external or third-party inspection is required. The inspector shall not be the work supervisor.

4.6.4.13 INSPECTION AND TEST AREAS

Each inspection and test area shall be in the order of 4 to 10 m² in size with the total of the inspection and test areas being approximately 20% of the total area to be painted.



4.6.4.14 INSPECTION REQUIREMENTS

The following inspections shall be carried out:

1. Initial inspection – in accordance with AS/NZS 2312.1:2014 Section 9.6.2
2. Surface preparation – in accordance with AS/NZS 2312.1:2014 Section 9.6.3
3. Paint application – in accordance with AS/NZS 2312.1:2014 Section 9.6.3
4. Final inspection – in accordance with AS/NZS 2312.1:2014 Section 9.6.5

4.6.4.15 RECORDS AND INSPECTION REPORT

The inspector shall keep detailed records and reports in accordance with the requirements of AS/NZS 2312.1:2014 Section 9.7.

4.6.4.16 COATING REPAIRS

Should any of the coating application work not comply with the provisions of this specification then the areas concerned shall be repaired to the satisfaction of the Principal. Such repair work may include removal of the coating, followed by surface preparation and application of a new coating. The procedure for any repair work shall be reviewed by the Principal. All repair works undertaken in accordance with this clause shall be performed at the Contractor's expense.

For the purpose of this clause, non-complying work shall include coating failure due to yellowing, loss of adhesion, and colour change during the contract liability period.

All areas of coatings damaged by the Contractor during its operation shall be made good to the satisfaction of the Principal.

4.6.4.17 TRIAL APPLICATION

Unless otherwise approved by the Principal a trial application on a test area of the actual substrate of not less than 1 m² or a test panel made from the same substrate shall be conducted fourteen (14) days prior to the commencement of coating work. The test area or test panel shall be prepared and coated by the Contractor to satisfy the requirements of this specification.

Actual coverage rates of the coating system shall be recorded, in order that due allowance may be made in the full scale application. Additional requirements or observations shall be recorded and considered for the full scale application.

If the coating trial application is deemed by the Principal not to comply with the requirements of this Specification a further coating application shall be made until the performance criteria of this specification are met.

In the event that the trial application is rejected, the Contractor shall remove and dispose of any work deemed as unacceptable by the Principal, submit a new proposal to rectify the deficiencies (including prequalification testing of any new materials/methods) and repeat the trial application as described above. Any delays caused through rejection shall not constitute justification for extension of time.

4.6.4.18 REQUIREMENTS FOR FUTURE MAINTENANCE OF COATINGS

The Contractor shall provide the manufacturer's recommendations as follows:

5. The methods of preparation to be used in the event that re-coating of the coated surface is required.
6. Which types of coating, other than the original product, are compatible with the finish coat for re-coating purposes.



7. The technique which can be used to repair local damage to the coating, with particular reference to colour and gloss matching of finish coats applied after a time lapse of 5 years.
8. The most appropriate techniques for cleaning of the finish coat to remove surface soiling, where possible, without damage to the existing finish.

4.6.5 Hot Dip Galvanizing to Above Ground Steelwork

All hot dip galvanising of structural steel shall be in accordance with this specification and the requirements of AS/NZS 2312.2.

The coating system shall be selected on the following basis:

- Atmospheric Corrosivity as defined in AS/NZS 2312.1
- Surface preparation to BS EN ISO 8501-1
- A minimum durability (years to first maintenance) of 25+ years in accordance with AS/NZS 2312.1 (unless specified otherwise – refer **Façade Schedules**).
- FRL requirement in accordance with the **Fire Engineer's Specification** (if relevant)

The Façade Contractor is to engage a suitable professional to:

- Assess the atmospheric corrosivity category related to the site location,
- Assess the durability requirements of the nominated building elements e.g. Determine whether steelwork is internal or external to the air barrier and,
- Select a suitable coating system and surface preparation.

Internal steelwork refers to steelwork internal to the air barrier. External steelwork refers to steelwork forming part of the air barrier or externally exposed steelwork.

4.6.5.1 FABRICATED STEELWORK

Hot dip galvanizing of structural steel shall be carried out after fabrication to AS 4680 by approved galvanizers in accordance with the relevant codes.

Site welding or cutting shall be repaired by suitable procedures to ensure equivalent corrosion resistance.

In-line galvanizing products shall not be substituted for after fabrication hot dip galvanizing. Work containing factory fabrication carried out after galvanizing or in-line galvanized components where hot dip galvanizing is required may be rejected in whole or in part.

Galvanizers shall be members of the relevant industry or governing association. If directed, submit galvanizer's certificate of compliance.

Galvanizing shall be continuous, smooth and evenly distributed, and free from defects including lumps, blisters, gritty or uncoated areas, spots, dross or flux.

Straighten steelwork distorted during the galvanizing process by approved procedures.

Handle galvanized components to avoid any mechanical damage to galvanized surfaces.

If paint coating onto galvanised steel refer **Façade Schedules**.

4.6.5.2 GALVANIZING REPAIR

Repair damage to non-visible galvanizing, including damage by welding, grinding or cutting, or other visible steel, with an approved zinc-rich primer (cold-galvanizing) with zinc content not less than 90% dry film weight, in accordance with the relevant codes. Submit details.



Reinstatement of galvanizing will only be approved to areas damaged during installation. Reinstatement of poor quality galvanizing will not be approved.

Provide sufficient coats of primer so dry film thickness is not less than the equivalent thickness of the adjacent hot-dip galvanizing. Each coat shall overlap the previous coat by 50 mm.

Clean damaged areas in accordance with relevant codes and the primer manufacturer's instructions. Unless otherwise required, prepare surface by mechanical or manual wire brushing.

Prevent dust contamination until primer is dry. Any repair showing signs of dust contamination shall be removed and replaced.

4.6.5.3 PASSIVATION

If required, provide chromate passivation treatment to visible galvanizing to minimise staining which may occur on articles unable to be stored in dry, well ventilated conditions.

Passivate galvanized surfaces to be cast into or in contact with concrete by dipping in 0.1% sodium dichromate solution.

4.6.6 Prefinished/Pre-painted Steel Sheet

All precoated or prepainted steel sheet shall be in accordance with this specification and the requirements of AS/NZS 2728 or AS 1397 as relevant.

The coating system shall be selected on the following basis:

- Atmospheric Corrosivity as defined in AS 1397 or AS/NZS 2728,
- For hot dip zinc coated steel, a minimum durability (years to first maintenance) of 25+ years in accordance with AS/NZS 2312.1 (unless specified otherwise – refer **Façade Schedules**).
- For prepainted steel sheet a minimum warranty period of 10 years (unless specified otherwise – refer **Façade Schedules**).
- FRL requirement in accordance with the **Fire Engineer's Specification** (if relevant)
- Compatibility of materials in contact including fasteners, washers, and water run-off from adjacent surfaces.

The design and installation of prefinished/prepainted steel sheet wall cladding shall conform to the requirements of AS 1562.1 and AS 4040.2.

4.6.6.1 TREATMENT OF CUT ENDS AND PENETRATIONS THROUGH STEEL SHEET

Steel sheet cladding is to be detailed such that cut ends are concealed such as terminated behind flashings or within the wall cavity.

If there are any visible cut edges, the Façade Contractor is to propose a treatment for protection against corrosion for approval by the Client or Clients' nominated representative.

Where screws penetrate wall sheeting, a sealing washer comprising EPDM is to be used.

4.6.6.2 SCREW FASTENERS

All fasteners for prefinished/pre-painted steel sheet shall be selected based on the following basis:

- Atmospheric Corrosivity as defined in AS 4312 (as per prefinished/pre-painted steel sheet requirement),
- Corrosion Resistance Class as defined in AS 3566.2. Unless noted otherwise (refer **Façade Schedules**), Class 4 is to be used.



- Minimum Durability (as per prefinished/pre-painted steel sheet requirement),
- Compatibility of materials in contact including washers, cladding and water run-off from adjacent surfaces.

4.6.7 Zinc Aluminium Coatings to Above Ground Steelwork

All zinc aluminium coatings shall be in accordance with the manufacturer specification and the requirements of AS 1397.

4.7 ZINCALUME[®] Steel

4.7.1 General

All ZINCALUME[®] sheeting shall comply with AS/NZS 1365:1996 (R2016), AS 1397:2021, and be supplied by an ISO 9001:2015 Quality System–certified manufacturer.

4.7.2 Cleats and Sleeves

Where sheets are fixed by means of cleats or brackets, these shall be manufactured from ZINCALUME sheet/6063-T6 anodised aluminium.

4.7.3 Soft Soldering/Welding of Zinc

Zinc shall not be joined by either welding or soft soldering processes.

4.7.4 Zinc Finishes

AM125 Aluminium / zinc / magnesium alloy coated

4.7.5 ZINCALUM Laser Cutting

The approval is needed from the ZINCALUM sheet fabricator, to confirm that laser cutting is an acceptable method for perforating the Zincalume. does not compromise the integrity of the AM125 coating when performed correctly through proper process control.

The followings shall be managed by the fabricator

- Correct calibration of the laser cutting equipment
- Appropriate distance between the laser head and the material
- Avoidance of overheating or burning during the cut

Fibre optic laser systems is recommended and Laser 3D shall be ISO 9001 certified, to ensure consistent quality management. The fabricator to coordinate directly with the supplier - BlueScope's warranty team to ensure all requirements are met.

4.8 Aluminium

Provide approved aluminium extrusions, plate and sheet of alloy and grades suitable for the structural application, applied finishes and installation conditions in accordance with the relevant Standards. All aluminium materials shall be of consistent high quality regardless of source.

Submit details including proposed alloy types with supporting justification data and source.



Manufacturers shall be approved established manufacturers with a reputation for producing high quality material with a minimum of 5 years documented continuous track record.

Mill finish on any surface, visible or non-visible surfaces will not be accepted.

Submit manufacturer's certificate of compliance or test report for each batch of aluminium supplied. Each batch shall be suitably identified and cross referenced with the certificate.

4.8.1 Aluminium Fabrication

Fabricate aluminium in accordance with approved shop drawings and prototypes. Fabrication tolerances shall be as per the **Façade Schedules**.

Cut edges, drilled holes, riveted joints and flat sheets shall be clean, neat, and free from burrs and indentations. Remove sharp edges without excessive radiusing.

4.8.2 Aluminium Sheet

Unless otherwise indicated or required, sheet alloy shall be Grade 5005, Temper H34 or Grade 6061, Temper T1 and be suitable for anodising.

4.8.3 Sub-Framing

Unless otherwise indicated or required, all external sub-framing shall be aluminium Grade 6063-T5 or approved equivalent.

4.8.4 Sheet thickness for external cladding

Unless otherwise indicated on the Drawings, thickness of sheet aluminium for external cladding shall be not less than 3 mm.

4.8.5 Extrusions

Unless otherwise indicated or required, extruded aluminium alloy shall be Grade 6063, Temper T5 or T6 and be suitable for anodising.

Extrusion shapes and thicknesses shall withstand all required loads, and shall be rigid with straight and sharply defined profiles, free of draw marks.

Extrusion walls and screw flutes shall be of sufficient thickness for all fixings including flush countersinking where required.

Extrusion wall thickness shall be not less than 2.5 mm for structural applications and not less than 1.5 mm for non-structural applications.

4.8.6 Flatness of Metal Panels

All visible flat exterior metal panels of the façade system (including back-pans of shadowboxes) shall be designed, fabricated, and installed in such a manner that they are visually flat when viewed from any angle. Any short length distortions, ripples, waves, oil canning, or telegraphing of fasteners shall not be permitted. Provisions shall be made to allow for differential thermal expansion between framing members and the exposed metal of the façade system without noise and without distortion of the exposed face.



Visual flatness shall be addressed across a range of temperatures, with panels detailed to allow for thermal expansion / contraction without buckling or warping of panels. This is to be addressed in design and where necessary through checking panel flatness under the specified range of temperatures.

In the event that metal flatness requires interpretation by measurement, this shall be done by measuring and calculating the slope between any two points on the exposed surface 25 mm apart. The slope shall not exceed 0.5% from the nominal plane of the surface, when measured at an ambient temperature of 30°C.

Visible backpans for shadowbox spandrels shall be powder coated aluminium not less than 3mm thickness. The thickness and fixing of the backpan shall be designed to accept the full design loads.

4.8.7 Joints and Junctions

All visible joints shall be fixed by concealed means, unless otherwise indicated on the Drawings or approved in writing.

Fit exposed joints accurately to provide close continuous contact to a fine hairline. Ensure continuity of finish colour and texture without surface variations at joints.

Make junctions with concealed mechanical connectors so that no fixings, pins, screws, pressure indentations and the like are visible on exposed faces.

Sections shall be sized to eliminate edge projection or misalignment at joints.

Where required, joints shall be watertight and weather tight.

Other than for concealed stud welding, aluminium extrusions shall not be welded, except where approved in writing, and, if approved, only on concealed surfaces.

Joints between fabricated assemblies may be concealed with suitable aluminium extruded covers or clip-ons.

4.8.8 Aluminium Finishes

Applied coatings shall not fail or deteriorate during the warranty period for the normal exposure of the installed work, including atmospheric conditions and ultra-violet light, including:

- Cracking, peeling, pitting or corrosion, or similar defects visible from 1500 mm distance.
- Fading or colour change greater than 5 Hunter Colour Difference Units measured in accordance with ASTM D2244 for products of US origin.
- Chalking greater than 8 numerical rating points for non-metallic colours, and 6 points for non-metallic whites, when measured in accordance with ASTM D4214.
- All applied coatings shall be checked for compatibility with adjacent materials, substrates and finishes.

4.8.8.1 COLOUR AND FINISH

Refer to Architect's Finishes Schedule and approved control samples for required colours, gloss, texture, patina and the like.

4.8.8.2 PACKAGING AND HANDLING OF COATED COMPONENTS

Do not handle or transport factory coated components until the applied coating is dry and properly cured to achieve maximum hardness.

All coated components shall be protected by approved methods to prevent abrasion and other damage to the surface finish during handling, transportation and until the completion of all construction processes. Submit details of the proposed methods.



Provide appropriate factory applied packaging and protective materials to all finished aluminium surfaces. Do not use materials liable to bond to the substrate when exposed to sunlight or weather.

4.8.8.3 QUALITY RECORDS AND TESTING

Submit production and test records for coating system. Indicate test procedures, acceptance criteria and pass/fail rate.

4.8.8.4 REPAIR PROCEDURES

Submit full details for cleaning, maintenance and repair procedures. Submit sample panels indicating before and after examples of repair in the factory and on site of typical damage types.

Damaged or distorted work delivered to the Site shall not be installed. Where finishes are damaged on site, including damage caused by unauthorised site cutting, drilling or the like, remove and replace the damaged item, or restore the surface to its original condition by approved procedures.

4.8.8.5 POWDER COAT FINISHES

All powder coating to be in accordance with AS 3715 and AAMA 2604/2605.

Where required, finish metals with an approved powder coating system by an approved manufacturer and applicator in accordance with the relevant Standards to match the approved samples. Submit details, including details of preparation and pre-treatment.

The powder coating system shall be a thermosetting polyester resin, applied electrostatically and oven cured in accordance with the manufacturer's recommendation.

Manufacturers of powder coat systems that may be acceptable are Interpon Powder Coatings (Akzo Nobel).

The minimum dry film thickness (DFT) shall be as per the manufacturer's recommendations.

The applicator is to be an approved Qualicoat applicator.

The Façade Contractor is required to provide written evidence that the applicator is approved by powder coat supplier and that the conditions of warranty provision will be satisfied.

4.8.8.6 ANODISE FINISHES

Where required, finish metals by an approved anodising coating process and anodiser in accordance with AS 1231 to match the approved samples. Submit details, including details of preparation and pre-treatment.

Unless otherwise noted, anodizing grade and average thickness shall be:

- External: 25 microns with local thickness not less than 20 microns (AA25).
- Internal: 15 microns with local thickness not less than 12 microns (AA15).
- The Anodiser is to provide test reports demonstrating anodising grade and thickness in accordance with specification requirements.

4.8.9 Prefinished Aluminium Sheet

Panels shall be PVDF coil coated with a coating which satisfies the following:

- Coating thickness not exceeding 1mm
- Spread of Flame index in accordance with AS1530.3 not greater than 0



4.9 Stainless Steel

Provide all stainless steel in the required profiles, finishes, sizes and grades.

Exposed stainless steel shall be to AS/NZS 4673 Grade 316. Grade 302 shall not be used. Grade 316L shall be used for welded applications. Structural applications and all exposed stainless steel works shall be Grade 316L with roughness index $R_a < 20$. Grade 304 shall only be used for concealed applications and only with prior approval from the Architect and Inhabit.

4.9.1 Mechanical Finish

Unless otherwise indicated on Drawings, exposed stainless steel shall be finished to match approved samples in accordance with AS/NZS 4673, designated as follows:

- Linished No.4 - Grain shall be vertical in all locations, unless otherwise indicated on the Drawings.

4.9.2 Flatness

Flatness of architectural stainless steel shall be not less than "stretcher level" grade. Submit manufacturer's specifications – with no oil canning.

4.9.3 Colour

Where indicated on the Drawings to be colour finished, stainless steel is to be colour treated by an approved permanent process to match approved colour samples. Submit details.

4.9.4 Certification

Submit certificate of compliance or test report and properly identify each batch of stainless steel in accordance with the relevant Standard.

4.10 Timber

All timber shall be compliant with AS 2796.1 and AS 2796.2 and any other relevant standards.

All timber products and materials used in permanent or temporary works shall be:

- Certified by either Australian Forrest Certification Scheme (AFCS), or
- Is to be from a reused source, or
- Is to be a combination of the two

If the timber proposed in AFCS certified timber, it must be accompanied by the Chain of Custody (CoC) certificate for all parties involved in the handling, transforming and manufacturing process that will take raw timber from the forest to site. CoC certificates are to comply with AS 4707.

These requirements apply to:

- Formwork and other temporary installations of timber (e.g. hoardings)
- Structural and non-structural timber, including internal walls, floors and roof structures
- External and internal cladding
- Flooring, wall, and ceiling finished; and
- Internal and external joinery, windows, doors, and other specialist uses of timber, such as installed furnishings or balustrades
- CoC certificates are required from each link in the supply chain, which includes:
 - Roundwood producers or traders



- Primary and secondary processors
- Manufacturers and re-manufacturers; and
- Traders and/or retailers who break up packages and resell goods in unit or a different form from its originally purchased form including the Façade Contractor

In terms of transport between the links, and transport or haulage operations will need to be included either within the CoC system of a link whether owned or contracted by an organisation or, if independent of a link, the transport or haulage organisation will need to be certified to this Standard.

Where practical all timbers shall be sized to minimise wastage and utilise standard sizes as in accordance with relevant Australian standards.

All elements to be cut from quarter sawn timber. Crown cut/back sawn timber will not be acceptable.

Aesthetically varying material (due to range of species etc.) is to be blended throughout the façade. The blending requirement is to be approved by the Architect.

Timber cladding boards and visible framing shall be free from decay, insect attack (including pinholes) and free from knots, splits and shakes on faces to be exposed.

Arris, splaying, margin and branched knots are not acceptable.

Resin pockets are not acceptable in any event on any exposed surface.

Exposed pith is not acceptable.

Metallic fixings and fixtures in contact with the timber window frames shall comply with relevant Australian Standards to ensure there is no corrosion of metals by wood.

Adequate separation shall be provided to protect any adjacent elements from corrosion precipitated by tannin or other emissions from selected timber species.

Any external horizontal timber elements shall be sloped to a minimum 5°

The Façade Contractor is required to comply with the following in respect to material selection and installation:

- Approval must be obtained from the Superintendent before substituting listed materials from the schedule; and
- Throughout construction, and upon completion of construction, the Façade Contractor must undertake an audit of listed materials to ensure the correct products have been used.

The Façade Contractor shall be responsible for providing the following documentation for Green Star interiors PILOT certification (if relevant):

- Short Report providing an as-installed timber schedule showing how the credit criteria are met. The schedule must provide descriptions of each timber use or timber product; quantity of timber and timber products; identifying whether the timber is reused, certified or not applicable; the source of reused timber; and the name of the forest certification scheme and reference to chain of custody certificates or invoiced containing chain of custody codes or serial numbers for certified timber; and
- Evidence of compliance for each timber item used during the project (including formwork). Evidence must be in the form of:
 - For re-used timber: receipts or documentation showing that the item is re-used; or
 - For new timber: chain of custody certificates, codes or serial numbers.



4.10.1 Timber for external use as framing, sunshades, fins, beads and cladding

Structural timber components shall be made from the heartwood of one of the species listed below:

- Australian White Mahogany
- Grey Ironbark
- Red Ironbark
- Coastal Blackbutt

The timber shall have the following performance requirements:

- A natural toughness measure of M (Medium) or H (High) in accordance with AS1684.2
- A natural durability class of heartwood of class 1 in accordance with AS1684.2
- A natural termite resistant of heartwood class R – Resistant to termite in accordance with AS1684.2
- Timber to be minimum structural grade F17
- Do not use sapwood
- Timber to have group rating of 3 or better (i.e. 1,2 or 3) to comply with BCA Specification C1.10a OR a spread of flame index not more than 9 and a smoke development index not more than 8 if the spread of flame index is more than 5, to comply with BCA Specification C1.10.

The Façade Contractor may propose an alternative hardwood for consideration by the Superintendent, which conforms to the performance requirements. Alternative hardwoods will need to be approved.

4.10.2 Plywood components

All plywood shall comply with AS/NZS 1604.3 and AS/NZS 2271 and shall be treated to H4, A-Bond, structural grade. Provide evidence of compliance with BCA Fire requirements

4.10.3 Adhesives

Finger joints and laminated members will not be permitted (unless noted otherwise in this Specification) unless specifically agreed in time of tender.

Wood adhesives shall not be used unless approved by the Superintendent. Where used wood adhesives shall be in accordance with relevant Australian Standards.

4.10.4 Testing and verification

4.10.4.1 SELECTION OF TIMBER PRODUCTS

Provide samples to the Superintendent of all timber species proposed for the Works.

4.10.4.2 DESIGN SAMPLES

Prior to ordering any materials, the Façade Contractor shall first produce a range of samples showing the expected variations in appearance.

The Façade Contractor shall produce up to 5 additional samples of each species without additional cost to the Contract Sum.

4.10.4.3 SIZE OF THE FAÇADE CONTRACTOR'S SAMPLES

The Façade Contractor's samples shall be section size x 1200mm



4.10.4.4 RANGE DEMONSTRATING APPROVED SAMPLE

Once the Architect has selected which Façade Contractor's sample satisfied its design / aesthetic requirements, the samples shall be kept and used as the quality verification benchmark of timber supplied to the project.

Saw these into four equal pieces. Label and sign each sample set. Retain one set and distribute the samples to the following parties:

- Main Contractor
- Superintendent
- Architect

4.11 Insulation

Insulation shall comply with AS/NZS 4859.1 and BS EN 13162. It shall be inert, durable, rot and vermin proof, CFC and HCFC free, not support mould fungal or bacteria growth. It shall be sufficiently robust and cohesive to allow it to be removed and replaced in service without loss of material or performance. It shall not be injurious to human health when being installed, in service or being removed or replaced.

Insulation materials shall be robust and durable under the conditions that they are to be use. If Insulation is provided in front of the weather/air seal line, it shall be durable and its thermal performance shall not degrade under exposed conditions. Unmodified Rockwool (mineral wool) is not considered durable under exposed conditions. Modified rockwool (mineral wool) maybe considered acceptable subject to submission of material test data to the satisfaction of the Architect and Superintendent.

All thermal insulation used in the project is to have zero Ozone Depleting Potential (ODP) in both manufacture and composition. Blowing agents (where used) shall also have an ODP of zero. Insulation shall not contain volatile organic components which may be subject to volatilisation and condensation on the inside of glass components of the façade system at the operating temperatures as specified in this document.

Insulation to be non-combustible in accordance with the NCC requirements as outlined in clause 3.9 Fire Performance.

4.12 Prohibited Materials

The following materials are not acceptable for use on this project:

- Non-compliant building materials: Materials that do not comply with the requirements of National Construction Code (NCC), other relevant laws and Australian Standards.
- Asbestos – materials including insulation, gaskets, coatings, GRC shall provide testing evidence in accordance with AS 4964
- Lead where the metal or its corrosion products may be directly ingested, inhaled or absorbed, or any lead-based paints or primers
- Chromate paint pigment or chromate water treatment
- Urea formaldehyde foam or materials which may release formaldehyde in quantities which may be hazardous or irritant
- Synthetic mineral fibres except in accordance with Work-Safe Australia National Standard and National Code of Practice
- Materials in which chlorofluorocarbons, hydro-chlorofluorocarbons or hexa-fluoroacetones have been used as a blowing agent
- Chlorofluorocarbons
- Polychlorinated biphenyls



- Other substances generally known to be deleterious to health or safety or which would adversely affect the works
- Paints, sealants, adhesives, flooring and carpets that contain unacceptable levels of Volatile Organic Compounds (VOCs).
- Engineered wood products that have unacceptable levels of formaldehyde emissions.

In instances where materials fall beyond the NCC and Australian Standards, ASTM standards are to be complied with.



5.0 Execution

5.1 Fabrication

5.1.1 Fixings to Aluminium

Fixings to aluminium or aluminium alloys shall be non-magnetic stainless steel unless otherwise indicated on the Drawings. Cadmium-plated steel or aluminium fixings shall not be used. Self-tapping screws shall be stainless steel. Non-visible screws may be pan-head type.

Visible fixings where required and/or approved, shall be indicated on the shop drawings. Visible fixings shall be finished to match adjacent substrate. Unless otherwise indicated on the Drawings, visible screws shall be countersunk stainless steel with Phillips or "pozidrive" heads, evenly and neatly located.

5.1.1.1 TESTING OF ALUMINIUM FIXINGS

Carry out testing of the tensile strength of welded aluminium stud fixings and extrusion flute connections by an approved testing programme. Submit details of proposed test programme.

Test programme shall include an initial proof testing procedure of not less than 100 examples of each type of fixing, and subsequent trend testing of not less than 10 examples progressive through the fabrication programme at not less than 8 evenly spaced intervals.

Approval of fixings shall require that the average of each set of 6 test failure loads be not less than 2x the design load, and absence of permanent deformation at 1.5x the design load.

5.1.2 Welded Studs

Where required for fixing face sheeting to support frames, provide not less than M6 fully threaded welded studs, of approved type, and welded by approved procedures. Submit complete details including samples, and pull out test results. Include preliminary and on-going production test results.

Welded studs must not distort exposed panel surfaces, or impact on the colour, uniformity and patina of the exposed coating system. Visual mock-ups and continuous monitoring is required by the Façade Contractor. The following panel thickness is required for stud fixings:

- Anodized and metallic painted panels require 4mm or greater solid aluminium
- Flat painted panels require 3mm or greater solid aluminium

5.1.3 Holes

Provide holes and connections for site assembly and to accommodate work of others as required. Holes shall be drilled, or punched and reamed, perpendicular to the surface.

5.1.4 Marking

Provide suitable clear marking to enable correct setting out, and installation. Marking shall be of a type that can be removed with water or solvents after assembly. Marking should be positioned on unexposed surfaces where possible.

5.1.5 Built-up Members and Reinforcement

Where two or more sections of aluminium are used in build-up members, contact surfaces shall be smooth, true and even, and secured so that the joints are tight without the use of filling materials.



Steel reinforcement of aluminium members shall be completely enclosed and separated from aluminium by approved methods. Submit details.

5.1.6 Moving Parts

Moving parts shall operate freely and smoothly, without binding or sticking, at correct tensions or operating forces, lubricated where appropriate.

5.1.7 Protection

Protect finished aluminium surfaces to prevent damage during transportation, storage, installation, and until the completion of all construction processes.

Provide factory applied protective film, tape or coatings which will not bond to the aluminium surfaces when exposed to sunlight or weather.

5.1.8 Welding of Steel

Carry out all welding in accordance with appearance and quality of approved sample welds, the Structural Steel Specification and relevant Standards.

Submit details including detailing of all joints, procedures, and correction of defective work. Do not commence until approved in writing.

Welded parts shall be accurately fabricated to ensure proper fit. All welding equipment shall be of suitable type and in good condition.

5.1.8.1 PROCEDURES

All welding shall be continuous unless otherwise approved. Tack welding or skip welding will not be approved where items are to be galvanized.

Other than site welds indicated on approved shop drawings, do not weld on site without prior approval. Where practical, locate site welds in positions for down hand welding.

Do not weld:

- Finished surfaces.
- Adjacent to finished surfaces, unless adequately protected from damage.

5.1.8.2 PREPARATION

Prepare fusion surfaces to AS/NZS 1554. Surfaces must be dry. Warm the surfaces if required to remove condensation.

Remove welding slag by chipping before depositing subsequent runs.

5.1.8.3 TACK WELDS

Tack welding may only be used with express approval. Tack welds to be minimum 50mm long.

5.1.8.4 BUTT WELDS

Butt welds shall be full penetration welds between prepared fusion faces, unless otherwise specified. Carry out back chipping, grinding or gouging of the deposited weld as required to obviate imperfections in the root run. Grind butt welds flush without loss of parent metal.



5.1.8.5 FILLET WELDS

Deposit fillet welds to the required length, throat thickness and with partial or full penetration as specified.

5.1.8.6 TEMPORARY ATTACHMENT

Do not weld temporary attachments to principle joints. Obtain approval of the position of welds for temporary attachments.

5.1.8.7 FINISH

All welds shall be descaled and free of surface and internal cracks, slag inclusion, and porosity.

Visible welds shall ground smooth to match approved samples. Non-visible welds shall be ground smooth to ensure uniform galvanizing if uneven or rough.

5.1.8.8 EXPERIENCE OF WELDERS

Welds shall be made only by welders who have previously been qualified for the type of work required. Submit certification.

5.1.8.9 WELD TESTING

If required, carry out weld testing by an independent testing laboratory. Submit test results. In the event of test failure, rectify the defect and repeat the test.

5.1.8.10 WELDING DISSIMILAR METALS

Submit details of welding of dissimilar metals, including:

- Type and thickness of materials to be welded.
- Proposed joint preparation and welding procedures.
- Proposed filler metal.
- Expected dilution (proportion of fused parent metal in the weld metal).

5.1.9 Connections

5.1.9.1 ANCHORAGES

Provide all required anchors and embedments to support the work from the base-structure, including proprietary anchorages, fabricated anchor bolts and other embedments as required with holding power of not less than 3x design load.

Select and install proprietary anchors suitable for the substrates and conditions in accordance with the manufacturer's recommendations. Submit details, including computations and shop drawings where required indicating type, size, depth, spacing and minimum edge clearances.

Do not use explosive shot fired devices unless approved in writing.

Anchors and embedments shall be hot dip galvanized steel or other approved durable non-corrosive finish.

Install anchors and embedments to tolerances that shall not exceed the values in the ***Façade Schedules***.

5.1.9.2 PROPRIETARY CHANNEL ANCHORAGES

Channel anchorages shall be approved proprietary types complete with polystyrene insets and plastic end caps. Submit details.



5.1.9.3 FABRICATED ANCHORS, HOLDING DOWN BOLTS AND EMBEDMENTS

Install anchorages in concrete in accordance with relevant Standards. Do not install anchors into post-tensioned concrete structures, void sections of precast concrete panels, and non-conventional concrete structures, unless approved in writing.

Refer to Drawings for concrete reinforcement locations, and position anchors to minimise risk of conflict with reinforcement.

Include sufficient tolerance to compensate for anchorages which become dislodged or damaged during placement of concrete.

Where required, supply anchors and embedments for installation by concrete subcontractor. Include templates, set-out drawings and installation instructions.

5.1.9.4 BRACKETS

Design and provide all required brackets in accordance with approved shop drawings and fix work to the base-structure so that all loads of the installed work are transmitted to the base structure through brackets.

There shall be no direct fixing or close contact between any part of the installed work and the base-structure or adjacent work, except through approved brackets, connections and anchors.

Brackets shall provide three-dimensional adjustment and accurate location of the work, and be rigidly fixed after the work is finally positioned.

Do not cut, bend or otherwise alter installed brackets without written approval. Any brackets modified without approval may be required to be removed and replaced.

Carry out and provide proof test results for all bracket types. Testing may be carried out in conjunction with anchor testing.

5.1.9.5 BRACKET TYPES

Provide dead load and restraint-only brackets as required. Indicated on shop drawings as follows:

- Dead load brackets shall be designed to support all loads imposed from any direction, without reliance on friction and slotted holes. Slotted holes will not be approved.
- Restraint brackets shall be designed to support only loads imposed perpendicular to the plane of the dead load (such as wind loads and service loads), and enable installed work to move as required to accommodate movements and deflections in the base-structure and the installed work due to any cause. Indicate on shop drawings methods for ensuring that restraint brackets do not transfer dead loads.

The design and finished installation shall enable all movement requirements to be capable of inspection and adjustment after installation.

Unless otherwise indicated on approved shop drawings, all site installed fixings and fasteners shall be capable of loosening and retightening.

All bracketry outside the air seal shall be stainless steel or powder-coated aluminium.

5.1.9.6 SHIMMING AND PACKING

Provide approved packing, spacers and shims as required to ensure proper installation in correct locations and to required tolerances. Material shall be of suitable durability and corrosion resistance for design life.



Structural shims shall be in accordance with AS 3850 where applicable.

The depth of packing shall not exceed maximum allowances indicated on shop drawings.

Packers shall be of adequate bearing area to transfer loads without inducing bending moments to bolts or studs and shall be fully engaged with the substrate over the full bearing area.

Do not use U-shaped shims or packers on sloped or vertical faces at dynamic connections or at static connections where there is any possibility of misalignment or loss of the shim or packer.

Temporary packers shall be easily visible, easily removable and capable of positive verification of removal. Remove progressively during sealing and grouting operations. Evidence of packers not removed will be sufficient cause for the rejection of related grouting or sealing, and all affected work may be required to be removed and replaced.

Any visible packers are to match the framing colour or colour approved by the Architect.

Packing material to be approved by the relevant building consultant / certifier with consideration to fire performance properties and in accordance with the NCC requirements as outlined in clause 3.9 Fire Performance.

5.1.10 Fixings

Provide all required fixings of approved types to assemble and install work in a neat, secure manner, including bolts, washers, screws, rivets, welds, proprietary fasteners, and the like. Include templates and other accessories for a complete installation. Submit details, including computations and shop drawings where required indicating type, size and spacing of all items.

Fixings shall be of the type and size appropriate for the purpose and location in accordance with manufacturer's instructions and shall be sufficient to transmit the imposed loads and stresses and ensure the rigidity of the assembly. All fixings shall be new and undamaged.

Do not use explosive shot fired devices unless approved in writing before commencing.

Fixings to finished work shall be concealed unless otherwise approved or indicated on the Drawings.

Fixings shall be installed to accommodate all substrate movements and thermal movement and enable generous on-site adjustment in accordance with approved shop drawings.

Items shall be delivered to site in durable packages, labelled to indicate type and intended use.

All fixings outside or through the air seal shall be stainless steel.

5.1.10.1 BOLTS, NUTS AND WASHERS:

All bolts, nuts and washers shall be galvanised High Strength Grade 8.8 to AS/NZS 1252, unless specified otherwise on the Engineer's drawings.

5.1.10.2 HSFG BOLTS:

High Strength Friction Grip Bolts and associate nuts and washers are to comply with AS 4100 and AS/NZS 1252 unless otherwise specified. Contact surfaces are to be left unpainted or prepared as otherwise indicated on the Engineer's drawings. HSFG Bolts are to be installed in accordance with AS 4100 and AS/NZS 1252 using a wax based lubricant and coronet type load indicator washers or an Engineer approved equivalent.



5.1.10.3 STAINLESS STEEL BOLTS, SCREWS, NUTS AND WASHERS:

Stainless steel bolts and nuts shall comply with AS/NZS 4673, strength A4, class 80. Stainless steel washers shall comply with AS/NZS 4673, Grade 316. Screws and fasteners are to be Grade 316 or equivalent.

5.1.10.4 CAST-IN PLATES AND CHANNELS:

Cast-in plates are to be Grade 250 with welded studs or mild steel reinforcement welded to the plates in accordance with the Engineer's drawings. Cast-in plates are to be hot dip galvanised after all the fitments have been attached.

Cast-in channels are to be hot-rolled, galvanised steel or an Engineer approved equivalent. They shall be designed and installed in accordance with the manufacturer's requirements. The channel shall be able to carry the design load including any prying forces at the most eccentric setting of the channel.

Cast-in channels shall be Halfen, Hilti, Jordahl or approved equivalent and be in accordance with SA TS 101.

5.1.10.5 HOLDING-DOWN BOLTS:

Holding-Down Bolts and associated nuts and washers shall be galvanised Grade 4.6 to AS/NZS 1252 unless otherwise specified. They shall be rigidly held in place at the top and bottom by tack welding to mild steel links before galvanising. They are to be located using a template that is firmly secured to the formwork or reinforcement.

The threaded length is to be sufficient to take up all construction tolerances, and is to be protected by taping and covering where impact damage may occur.

5.1.10.6 STUDS:

Shear studs shall be Nelson Studs or an Engineer approved equivalent welded to the structural steel in accordance with manufacturer requirements.

5.1.10.7 MASONRY ANCHORS:

All anchors are to be proprietary anchors with adequate test data and shall be installed in accordance with the manufacturer's requirements. The Façade Contractor is responsible for confirming that all edge distance, spacing and embedment requirements are satisfied.

Chemical anchors are not to be used where a fire-resistant fixing is required. Unless approved for such use by the manufacturer. Written confirmation is required.

Masonry anchors into both solid and hollow block are to be purpose designed anchors suitable for this purpose.

The Façade Contractor shall be responsible for ensuring that where masonry anchors clash with reinforcement, there is an alternative anchor set-out that satisfies the design requirements. End plate modifications to suit a new anchor set-out are to be carried out in accordance with AS/NZ3600, slotting of holes and flame cutting are not permitted. Holes in the concrete are to be repaired by dry packing with a 40MPa cementitious non-shrink grout.

5.1.11 Joints and Junctions

5.1.11.1 SEALANT MATERIALS

Provide approved elastomeric joint sealants and joint-fillers where indicated on the Drawings and required.



Sealant type and installation method shall be in accordance with the manufacturer's recommendations suitable to the location, function, substrates and performance requirements including movement, air infiltration, fire and acoustic requirements.

Submit complete details, specifications, installation and curing instructions, including manufacturer's certification that the proposed product selections are suitable for the joint types and are compatible with, non-staining, and non-bleeding to adjacent materials.

All sealants for each joint type shall be the same product and manufacturer.

Sealant fillers for floor surfaces shall be self-levelling.

5.1.11.2 MOVEMENT REQUIREMENTS

Where sealants are used to seal movement joints, movement capability of sealant shall be appropriate to expected maximum deflection or movement.

Unless otherwise approved, sealant depth for movement joints shall not exceed half the joint width.

5.1.11.3 FIRE RATED REQUIREMENTS

Provide approved fire-rated sealant and joint-fillers where indicated or required. Submit test results indicating fire rating and compliance with statutory requirements.

5.1.11.4 COMPATIBILITY

All proposed sealants and accessories shall be compatible with substrates, adjacent materials, accessories and other sealants, and shall be non-staining and non-bleeding.

Silicone sealants shall not be used adjacent to stone materials unless approved in writing.

5.1.11.5 COLOURED SEALANTS

Provide coloured or paintable sealants where required in accordance with approved samples. Sealants shall be resistant to staining due to dirt or pollution, and shall be able to be cleaned.

5.1.11.6 SEALANT ACCESSORIES

Provide all required accessories recommended by sealant manufacturer, including backing rods, bond breaker tape and the like. Submit details.

5.1.11.7 MANUFACTURER'S STANDARD TEST DATA

Submit manufacturer's standard test reports and certificates for testing previously performed on proposed sealants.

Include adhesion characteristics for all relevant surfaces including applied coatings to concrete, aluminium, glass and adjacent sealants and gaskets.

Include statement of differences between the proposed sealants and previously tested products, if any.

Include long-term aged performance or accelerated exposure performance for the above characteristics.

5.1.11.8 QUALITY RECORDS

Submit production and test records for sealants. Indicate test standards and procedures, acceptance criteria and pass/fail rate.



5.1.11.9 WARRANTY

All sealants shall be warranted for the Warranty Period for failure of, or due to, the following:

- Cohesion and adhesion.
- Change in hardness by more than ± 5 points on the Shore A durometer scale during the service life of the work.
- Fade or change in colour by more than five Hunter Colour Difference Units as measured by any spectrometer or colour-meter in accordance with ASTM D 2244 on an exposed coated surface which has been cleaned of external deposits with clear water and a soft cloth.
- Stain due to corrosion or bleeding of sealant.

5.1.11.10 STOCKPILING AND IDENTIFICATION

The sealant manufacturer shall stockpile sufficient material to ensure continuity of supply and compliance with approved manufacturer's specifications. All sealants shall be "batch numbered". Submit details of batch numbering records cross referenced to installation records.

5.1.11.11 SEALANT INSTALLATION

Comply with the sealant manufacturer's recommendations regarding surface preparation, priming, shelf-life, sealant bead application, and the acceptable range in surface temperature at time of application and for a period at least eight hours following sealant application.

Joints shall be prepared and sealed on the same working day.

5.1.11.12 CLEANING

Clean joint surfaces immediately before installation of backing rod and again before applying the sealant as recommended by sealant manufacturer. Remove all foreign matter such as dust, oil, grease, dirt, laitance, insecure coatings, moisture and other substances which could interfere with bond of sealant.

Remove protective tape or removable films and ensure that no residue remains.

Joint areas to be protected with masking shall be cleaned before application of tape or film.

Aluminium:

Cleaning compounds shall be applied with clean lint-free cloths. A two-wipe method of application shall be employed, where one cloth is used to wipe the surface dry and clean, and the second is used to apply the cleaning compound. The cleaning compound shall not be allowed to air-dry on the substrate.

Porous substrates, concrete, masonry:

Clean where necessary by grinding, mechanical abrasion, detergent washing or a combination of methods to ensure a clean sound interface.

Remove laitance mechanically. Remove formwork oils by blast cleaning. Remove loose particles that remain in joints following mechanical surface preparation by blasting with oil-free compressed air.

Clean and etch concrete and masonry joint surfaces as recommended by sealant manufacturer.

5.1.11.13 PRIMING, ETCHING AND SEALING

Prime or seal joint surfaces where recommended by sealant manufacturer. Do not allow primer/sealer to spill or migrate onto adjoining surfaces. Areas adjacent to joints to be sealed shall be protected where there is a likelihood that contamination by cleaning compound, primer or sealant could occur.



If recommended by the sealant manufacturer, roughen surfaces to remove protective coatings or imperfections which may prohibit provision of clean, sound base surface for sealant adhesion.

5.1.11.14 INSTALLATION OF SEALANT ACCESSORIES

Backer rod:

Install sealant backer rod for liquid elastomeric sealants, except where otherwise indicated, or not recommended by sealant manufacturer, at a proper depth to provide sealant bead profiles indicated on approved shop drawings. Backer material shall be of appropriate size and shape and shall be compressed between 25% and 50% when installed.

5.1.11.15 BOND BREAKING TAPE:

Install bond breaker tape where indicated and where required by the manufacturer's recommendations to ensure the proper performance of elastomeric sealants.

5.1.11.16 SHELF LIFE

Ensure that remaining shelf life of products is well within the likely installation period. Confirm expiration dates with manufacturer and scheduled installation dates with the Main Contractor.

5.1.11.17 CURING

Provide for the appropriate cure conditions, in accordance with the sealant manufacturer's written recommendations at factory and on-site. Protect external sealants from inclement weather until fully cured.

Cure sealants in compliance with manufacturer's recommendations, to obtain high early bond strength, internal cohesive strength and surface durability.

Submit details of procedures required for cure and protection of joint sealers during construction period to prevent deterioration or damage (other than normal wear and weathering) at time of Completion.

Do not relocate sealed components within the factory or on-site, until the joint has developed sufficient bond strength and cohesive integrity.

Installation of sealants between interior faces of windows and adjacent façade system shall be carried out at the same time using the same or compatible materials as the interior sealant installation for the façade system.

5.1.11.18 INSTALLATION OF SEALANTS

Install sealant to depths as recommended by the sealant manufacturer and indicated on the approved shop drawings. Install sealants during ambient temperature and humidity conditions recommended by the manufacturer.

Employ only proven installation techniques which will ensure that sealants are deposited in uniform, continuous ribbons without gaps or air pockets, and with complete "wetting" of joint surfaces equally on opposite sides.

Unless otherwise indicated on the approved shop drawings, fill vertical joints to a slightly concave surface, slightly below adjoining surfaces, and fill horizontal joints to slightly convex profile, so that joint will not trap moisture and dirt.

Apply sealant under pressure using a hand or power actuated gun or other appropriate means. The application tools shall have nozzles or size and shape as necessary to ensure that sealant beads are formed to the designed profiles. The surface of each joint shall be tooled so as to ensure the formation of a compacted bead,



free from cavities and flow irregularities. A lubricant (tooling aid) shall not be used unless the use of such a compound is expressly approved in writing by the relevant manufacturer.

Remove excess sealant and leave surfaces neat, smooth and clean, without smears on surrounding work. Tool joints where recommended by manufacturer or required. Remove all cartons and debris from site as the work progresses.

Remove any masking material immediately after application of sealant bead is complete and "clean-down" adjacent surfaces as work progresses. All finished work shall be left in a neat clean condition.

Provide weep holes where indicated or required.

5.1.11.19 FLASHINGS AND TRIMS

Provide all required flashings, baffles, trims, cappings and the like to prevent the entry of water and weather, and make neat and clean junctions with the base-structure and adjoining work. Where visible, provide matching materials and finishes. Include all fixings and sealing.

All flashings shall be cut and folded to approved profiles out of non-corrosive materials, with protective coatings as required. Flashing shall be factory fabricated in long lengths where practical, and pre-painted on visible surfaces. Submit details.

Flashing shall be of adequate stiffness to retain shape and to resist lifting by the wind. Make provision for differential movements and for separation of dissimilar materials.

Continuous flashings shall be welded or mechanically fixed to form continuous uninterrupted lengths. Corners shall be accurately scribed and mitred.

Where flashings are fitted to pre-formed rebates, coordinate cast-in grooves or reglets as required.

Flashings shall be of approved non-corrosive materials and profiles, fixed to accommodate differential movements and provide for separation of dissimilar materials. Submit details.

Construct weepholes as required to enable the passage of moisture to the outside of the building. Lap and seal all flashings by approved methods. Submit details.

Minimum sheet metal thickness for roof flashings and cappings shall be in accordance with AS 1562.2.

5.1.11.20 TEMPORARY FLASHING

Temporary flashing is to be installed in a manner that keeps all run-off away from other façade system materials and finishes. Run-off is to be directed to pavement level.

5.1.11.21 STRUCTURAL GLAZING ON-SITE.

Site structural glazing works are not permitted unless otherwise agreed with the project team. Approval will not be given for structural glazing on-site unless a QA and ITP plan is provided by the Façade Contractor that includes:

Methodology statement

- Product specification, batch and storage recording
- Substrate adhesion and compatibility verification
- Cleaning and priming requirements/procedures
- Application temperature and humidity requirements



- Temporary restraint requirements ensuring undisturbed curing and corresponding touch-up/make good procedures
- Sealant test procedures in accordance with manufacturer's requirements
- The manufacturer certification of glaziers in terms of suitability and experience

The ability to achieve the required joint under site conditions is to be proven prior to full site works commencing.

The Façade Contractor will be responsible for engaging and ensuring a representative from the structural silicone supplier is involved in the generation of QA and ITP plan and attends site to inspect glazing and deglazing works to confirm warranty provision. A letter is to be provided from the structural silicone supplier stating that they received design of structural silicone joints, witnessed fabrication, QA testing, audited QA records and procedures.

5.1.12 Openings

5.1.12.1 OPENINGS AND OPERABLE DOORS

Provide approved surround-frames where indicated, including all trim, fixings, seals and accessories. Submit details.

Where indicated on the Drawings, doors shall be mounted by:

- Hinges (Generally)
- Pivoting (Operable Louvre Doors)

Where required, hinge mechanisms shall be an approved proprietary type. Products which may be approved include those manufactured by Dorma, or equivalent. Coordinate with mechanical door operator suppliers where required.

Exposed fixings shall be stainless steel Grade 316. Door and surround-frame construction generally shall include all required internal reinforcement and exposed surfaces of:

- Aluminium.
- Stainless steel.

Refer to Drawings for required coatings and finishes. Fixings and concealed metalwork shall be of non-corrosive materials or coated to prevent corrosion.

Coordinate required recesses and cores in base-structure, grout in components where required, and make good. Coordinate the correct position and level of floor mats assemblies by others.

Refer to Architect's Finishes Schedule for finishes.

5.1.12.2 LOADS

Doors, opening sashes, and surround-frames shall be reinforced as required for wind loads and concentrated loads at hinges, closer and latch connections. Submit computations and indicate methods of protecting doors and opening sashes from damage when opened in high wind gust conditions. Include details of required closer types and actions.

5.1.12.3 FIRE RATING AND INSULATION

Refer Architectural and Fire Engineer's Specifications for fire rating and insulation requirements.



5.1.12.4 WEATHER SEALS AND ACCESSORIES

Provide all opening sashes with approved weather seals.

5.1.12.5 WARRANTY

Provide warranty for all doors and accessories, including provision for wind damage, for the Warranty Period.

5.1.12.6 HARDWARE

Supply all required door hardware and furniture of approved type and manufacture, suitable for the function and the intended use, in accordance with all relevant Standards and requirements (if any) of Statutory Authorities, and with approved samples and product data. Submit details.

Hardware and furniture may include locksets, latchsets, keys, hinges, closers, stops, buffers, push / pull bars and the like, and all fixings as indicated on the Drawings and Door Schedule.

Include all associated sundries including fixings, striker plates, shims, escutcheons and the like required for a complete installation, whether or not indicated.

Refer Architect's Door / Hardware Schedules.

Supply hardware properly packed and labelled in the manufacturer's original and unopened packets, including complete installation instructions with detailed delivery dockets.

If required, provide a system of construction locking.

Where practical mount hardware flush with the face of the frame with concealed fixings.

Unless otherwise indicated, opening units shall be fitted with approved locking devices.

5.1.12.7 SUBSTITUTIONS

Where scheduled hardware is available, no proposals for substitution will be considered unless otherwise approved.

Where scheduled hardware is not available for any reason, notify reason for unavailability and submit proposal for alternative products.

5.1.12.8 ESCAPE

Unless otherwise indicated on the Drawings, all door hardware shall enable escape from the inside of the room or area contained by the door hardware, with the exception of the roof where means of escape is into the plant room.

5.1.12.9 HANDING AND MOUNTING

5.1.12.10 HEIGHT

Verify correct handing of all items before supply.

All door furniture shall be suitable for mounting 1050 mm above finish floor level unless otherwise indicated.

5.1.12.11 KEY RINGS AND CABINET

Unless otherwise indicated, provide each key identified by a purpose made plastic or stamped metal label attached by a metal ring, and an approved proprietary lockable key cabinet for all keys collated and placed on separate compartments, pockets or hooks. Submit details.



Fit with construction cylinders for replacement with Abloy/Lockwood master key system by others.

5.1.13 Louvres and Grillage

Provide approved proprietary louvre assemblies, complete with all accessories and fixings, together with structural calculations for blades, frames, fixings, and blanked off sections. Submit drawings and details including structural and test data.

Design and/or select external louvre assemblies to withstand the required design wind pressure without failure or permanent distortion of members, and without blade flutter. Structural frames and stiffening rib members shall not deflect by an amount greater than span/360 when tested to the required design wind pressure.

Where required for access, provide framed and pivoted louvred access doors to match adjacent louvres.

Install louvres into frames by methods that resist unauthorised removal.

5.1.13.1 AIR FLOW

Design louvre assemblies to have a free area to meet the requirements specified by the Mechanical Engineer and to allow for free air flow requirements indicated on the Drawings. Submit verification of compliance with free-air movement and maximum pressure drop requirements.

Submit manufacturer's test reports, including previous tests of identical louvre design, or tests of the actual louvre to be used or a representative prototype.

5.1.13.2 WEATHERPROOF MECHANICAL LOUVRES

Mechanical louvre systems shall be of the horizontal blade type suitable for use where a high degree of weatherproofing is required. Refer to Mechanical Engineers Specification and Drawings for louvre panel layouts.

Louvre types shall be matched to mechanical plant requirements as per the Mechanical Engineer's documentation and Architect's specification.

5.1.13.3 MATERIALS AND FABRICATION

Mechanical louvres shall be manufactured using minimum 1.8mm thick extruded aluminium blades, suitably stiffened, in an extruded aluminium frame.

The mechanical louvre frame shall be of the standard flanged type or alternatively, channel type for flush fitting. The louvre frame shall be constructed with neatly mitred corners mechanically fixed for extra strength.

5.1.13.4 PERFORMANCE

The finished louvres shall achieve the required performance requirements e.g. required air flow and maximum pressure drop as noted in the ***Mechanical Engineer's Specification***

Blanking panels, where required, shall be fabricated using minimum 2mm thick powder coated aluminium sheet.

5.1.13.5 WEATHER PROTECTION AND DRAINAGE

Design and install internal mechanical louvre assemblies to be waterproof and weatherproof where indicated by Mechanical Engineer's drawings.



Unless otherwise approved, internal louvres or louvres subject to wind and weather shall incorporate two stage blade design.

Louvre assemblies shall be self-draining with internal drainage channels and outlets as required. Provide stainless steel or aluminium drip trays as required and where indicated on Drawings.

5.1.13.6 VERMIN MESH

Provide approved concealed grade 316 stainless steel bird wire mesh to the interior of all external louvre assemblies, and blank off plates to all other false louvres. Mesh is to be a minimum of 1.2mm diameter with 12mm x 12mm grid.

5.1.13.7 INSTALLATION AND REMOVAL

Louvres to plant rooms shall be designed to be removed and re-installed if required, for the removal/replacement of major plant items. The Façade Contractor is to demonstrate that panels that are designated to be demountable can be easily removed and reinstalled without damaging surrounding panels.

5.1.14 Moisture Resistant Anti-Carbonation Coating

All concrete surfaces and block walls behind the façade system, under the metal roof and in the open shafts to the building require membrane protection. Water Proof Membrane (WPM) references are listed.

All concrete surfaces (slabs and walls) that are not protected by an air-sealed façade system shall receive a protective finish layer. These surfaces are typically behind the rain-screen cladding and louvres.

5.1.14.1 DIFFUSION RESISTANCE

Carbon Dioxide Diffusion resistance: the anti-carbonation coating shall have a minimum equivalent Air Layer thickness $R > 100\text{m}$.

Moisture Vapour Diffusion resistance: the anti-carbonation coating shall have a maximum equivalent Air Layer thickness $SD < 2\text{m}$.

5.1.14.2 PENETRATION AND RE-INSTATEMENT

Where integrity of membrane is compromised by cuts, fixings, penetrations, holes, impact or any other detrimental action, membrane shall be re-applied and re-instated to manufacturer's specifications to ensure the original watertight integrity is retained to the primary structure and all other coated surfaces.

5.1.14.3 PAINTING

Membrane shall be painted matt black behind all louvre panels with an architecturally approved paint product prior to installation of louvres.

5.2 Installation

Install the façade system in accordance with shop drawings and prototypes to comply with all performance requirements, Codes and Standards and the requirements of the relevant Statutory Authorities.

Install the façade system plumb, level and true to line within required tolerances, and suitably anchored to the base-structure.



5.3 Delivery, Handling & Storage

All materials shall be packaged and palleted for safe and convenient handling by fork-lift vehicles and the Main Contractor's crane and hoisting equipment being used on the site.

The Façade Contractor is to ensure that all stillages, lifting beams, and other installation equipment is fully engineered, load limits are clearly stated and this equipment is inspected in accordance with Worksafe and other statutory requirements before this equipment is able to be used on the project.

Provide all required packing, wrapping and handling techniques to ensure that all panels are installed and handed over without any damage to the finished panel faces. Any damage to surface finishes may be ground for rejection and replacement.

Prefabricated panels stored in the factory or on site shall be stored under cover and not exposed to the weather. In the event that prefabricated panels cannot be stored under cover, they shall be stored in their stillages and shall be wrapped to prevent water ingress.

5.4 Project Conditions

5.4.1 Survey and Set-Out

Allow for engaging an experienced licensed surveyor to progressively verify actual locations and tolerances of profiles, openings, embedments and defects in base-structure, and make adjustments before fabrication and installation. Also address building movements that occur during construction in setting out the façade system works.

5.4.2 Installation Requirements & Tolerances

Commencing installation will be construed as complete acceptance of substrates and site conditions. Submit certification of acceptance of substrates and conditions, including the provision of reference lines and marks, and making specific reference to the type, number and location of the built-in anchors provided in the base-structure for the installation.

5.4.3 Post Fixed Concrete Anchors

All post-fixed concrete anchors are to be installed by an AEFAC Certified Installer.

5.4.4 Unitised Curtain Wall Gutters

Curtain wall gutters are to be kept clean and all foreign material, slurry, concrete dust and other building waste materials must be removed immediately prior to the installation of the row of curtain wall panels to the level above.

5.4.5 Firestops and Smoke Seals

Fire stops and smoke seals must provide continuous fire separation between fire compartments and floors. They are required to be installed on a floor-by-floor basis prior the zone between the structure, back walls, beams and columns being closed off.

5.4.6 Trims

Install all required flashings, trim, and seals to ensure the finished work is weatherproof and waterproof.

Do not fix through top of cappings unless noted otherwise.



5.4.7 Site Modifications

Obtain approval for site modifications before commencing modifications. Submit details.

Finished work which contains unauthorised site modifications, or work not in accordance with the approved shop drawings, may be required to be removed and replaced.

Unauthorised work may be approved subject to additional computations and testing at the Architect's sole discretion. If requested, carry out and pay for all such testing, and submit all computations, test results and any other information required.

5.4.8 Staged Completion

Façade system shall be installed in an orderly sequence. Refer master programme.

Provide temporary flashings at completion of each stage to waterproof and weatherproof the enclosed work. Remove temporary flashing before proceeding with subsequent work.

Sections of façade system which are left open for any reason shall be designed in accordance with the design wind serviceability load with an allowance for wind funnelling through the base-structure.

5.4.9 Hardware

Install all hardware and accessories including but not limited to latches, locks, openers and remote controllers.

Opening units shall be checked to ensure that operation is full, free and smooth, and that all operable hardware, locks and controllers are operating properly and smoothly.

Immediately prior to commissioning, complete site ITP checks for each operable façade component and check/adjust hardware as necessary to ensure proper operable component function upon practical completion. Take and maintain formal records of these checks and adjustments.

5.4.10 Temporary Marking

Provide temporary marking of glass (if necessary). Use a soluble marking compound and remove all traces on completion. Do not use lime or advertising stickers.

5.4.11 Drainage

Construct joint system and associated flashings to ensure that all moisture and condensation which penetrates the joint system is drained and discharged to the outside face of the façade system.

Ensure that drainage paths are continuous.

Ensure that weep-holes are located to prevent staining of finished coated surfaces.

5.4.12 Sub-Framing By Others

Where sub-framing has been installed by others inspect the sub-framing before fixing any cladding to ensure that sub-framing is plumb, level and aligned with required tolerances. Any defects which require remedial work shall be brought to the attention of the Main Contractor. No condition of the sub-framing shall provide grounds for voiding or limiting the required warranty.

Submit certificate by licensed surveyor to confirm that base framing is in the correct locations and to the required tolerances prior to commencement of installation.



5.4.13 Vapour Barrier

Where required, install a cavity membrane / vapour barrier to face of sub-framing in accordance with approved shop drawings.

Seal all penetrations and through fixings, and ensure continuous barrier to prevent passage of moisture to the inside of the façade system.

5.4.14 Drainage at Soffits

At the bottom of every externally sealed façade system surface (at the bottom of the lowest panel course and at each wall offset) provide for drainage at every panel joint. The flashing system shall drain all cavity moisture to the exterior.

Flashing shall be continuous, of stainless steel purpose-made to a stepped profile, fixed securely to the base-structure, with the top chased into a purpose formed rebate and sealed with polyurethane sealant. The flashing shall not protrude past the outside face of the façade system.

5.5 Completion

5.5.1 Progressive Cleaning

Clean the work progressively and daily by removing debris, waste, excess materials and the like from the work area.

5.5.2 Final Cleaning

At completion of installation, clean the work area thoroughly and clean the finished work to remove all marks, soiling and the like.

Finished work shall be free from defects and mechanical imperfections such as scratches, scrapes, dents, and abrasion.

5.5.3 Protection

At completion of installation, cover and protect finished work as required to prevent soiling, damage, or wear and tear during subsequent construction activities. Otherwise, clean and maintain finished work as frequently as necessary until the hand-over of the façade system. Protection shall be designed for removal without damage to finished surfaces. If requested, submit details of proposed protection materials and procedures.

5.5.4 Adjacent Work by Others

At the completion of adjacent work by others, including services work, attend the site, inspect finished work, and repair or replace damaged or dislodged work, and complete or make good finishing, trimming and sealing.

5.5.5 Responsibility

Be responsible for all damage and breakage of finished work until hand-over.

The Façade Contractor may assign responsibility for damage to another party where applicable, unless damage resulted from incorrect or unsuitable materials or installation procedures.



A Appendix A - Reference Standards

Works are to be in accordance with all applicable building codes and standards, together with any further state and municipal requirements.

The codes, standards and technical publications referenced below and elsewhere in this specification shall be the current edition (including the latest amendments). No substitutions to standards will be permitted without endorsement of the Client or Client's nominated representative.

The following abbreviations are used in this Specification:

AS	Australian Standards	CIRIA	Construction Industry Research and Information Association
AS/NZ	Australian/New Zealand Standard	NAAMM	National Association of Architectural Metal Manufacturers
BS	British Standard	EOTA	European Organisation for Technical Assessment
EN	European Norm	AWES	Australasian Wind Engineering Society
ASTM	American Society for Testing and Materials	ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineering
AAMA	American Architectural Manufacturers Association	CPNI	Centre for the Protection of National Infrastructure
CCAA	Cement Concrete & Aggregates Australia	AWS	American Welding Society
DIN	German Institute for Standardisation	CWCT	Centre for Window and Cladding Technology
GANA	Glass Association of North America	ANSI	American National Standards Institute
ISO	International Standards Association	AISC	American Institute of Steel Construction
GAA	Galvanizers Association of Australia	SCI	Steel Construction Institute
SSPC	The Society For Protective Coatings		

A.1 Loadings

Reference	Standard
AS/NZS 1170.0	Structural Design Actions General principles
AS/NZS 1170.1	Minimum Design Loads on Structures – Dead and Live Loads and Load Combinations
AS/NZS 1170.2	Minimum Design Loads on Structures – Wind Loads
AS1170.3	Minimum Design Loads on Structures – Snow Loads
AS1170.4	Structural Design Actions – Earthquake Actions in Australia
AWES-HB-001	Wind Loading Handbook for Australia & New Zealand



A.2 Glass

Reference	Standard
AS 1288	Glass in Buildings – Selection and Installation
AS/NZS 2208	Safety Glazing Materials in Buildings
AS/NZS 4667	Quality Requirements For Cut-To-Size And Processed Glass
ASTM C158	Standard test methods for strength of glass by flexure (determination of modulus of rupture)
ASTM C162	Standard terminology of glass and glass products.
ASTM C336	Standard test method for annealing point and strain point of glass by fibre elongation
ASTM C598	Standard test method of annealing point and strain point of glass by beam bending
ASTM C1036	Standard Specification for Flat Glass
ASTM E2431	Standard practice for determining the resistance of single glazed annealed architectural flat glass to thermal loadings.
BS EN 1063	Glass in building. Security glazing. Testing and classification of resistance against bullet attack
BS EN 356	Glass in building. Security glazing. Testing and classification of resistance against manual attack
BS 5516	Code of Practice for Design and Installation of Sloping and Vertical Patent Glazing
BS 5544	Specification for Anti-bandit Glazing
BS 6262	Code of Practice for Glazing for Buildings
BS 6375	Performance of Windows and Doors
DIN 18008	Glass in Building-Design and construction rules
EN 12600	Glass in Building- Pendulum Test- Impact Test Method and Classification
GANA	Flat Glass Manufacturers Association Guides
BS EN 16612	Glass in Building – Determination of the lateral load resistance of glass panels by calculation
NZS 4223	Code of practice for glazing in buildings
NZS 4223.1	Part 1: Glass selection and glazing
NZS 4223.3	Part 3: Human impact safety requirements
NZS 4223.4	Part 4: Wind, dead snow and live actions

A.3 Glass Coatings

Reference	Standard
DIN EN 1096	Glass in building – coated glass
DIN EN 410	Glass in building – determination of luminous and solar characteristics of glazing

A.4 Glass – Heat Treated

Reference	Standard
BS 6206	Specification for Impact Performance Requirements for Flat Safety Glass and Safety Plastics for Use in Building



BS EN 1863	Glass in Building – Heat Strengthened Soda Lime Silicate Glass
BS EN 12150	Glass in Building – Thermally Toughened Soda Lime Silicate Safety Glass
BS EN 14179	Glass in Building – Heat soaked thermally toughened soda lime silicate safety glass

A.5 Glass – Laminated

Reference	Standard
ASTM C1172	Standard Specification For Laminated Architectural Flat Glass.
ASTM D4501	Standard Test Method For Shear Strength Of Adhesive Bonds Between Rigid Substrates By The Block-Shear Method.
ASTM F521	Standard Test Methods For Bond Integrity Of Transparent Laminates
ASTM F942	Standard Guide For Selection Of Test Methods For Interlayer Materials For Aerospace Transparent Enclosures.
ISO 12543	Glass In Building – Laminated Glass And Laminated Safety Glass.

A.6 Glass – Insulating Glass Units

Reference	Standard
AS 4666	Insulated Glass Unit – Manufacturing And Testing
BS 5713	Specification For Hermetically Sealed Flat Double Glazed Units
ASTM E546	Standard Test Method for Frost/Dew Point of Sealed Insulating Glass Units
ASTM E2188	Standard test method for insulating glass unit performance
ASTM E2189	Standard test method for testing resistance to fogging in insulating glass unit
ASTM E2190	Standard specification for insulating glass unit performance
ASTM E2649	Standard test method for determining argon concentration in sealed insulating glass units using spart emission spectroscopy
EN 1279	Glass in Building – Insulating Glass Units
EN 1279-2	Part 2: Long term test method and requirements for moisture penetration
EN 1279-3	Part 3: Long term test method and requirements for gas leakage rate and gas concentration tolerances
EN 1279-6	Part 6: Factory production control and periodic tests
ISO 20492	Glass in buildings – Insulating glass
ISO 20492-1	Part 1: Durability of edge seals by climate tests
ISO 20492-2	Part 2: Chemical fogging tests
ISO 20492-3	Part 3: Gas concentration and gas leakage
ISO 20492-4	Part 4: Methods of test for the physical attributes of edge seals
NZS 4223.2	Code of Practice for glazing in buildings - Part 2: Insulating Glass Units

A.7 Glass – Bent/Curved

Reference	Standard
ASTM C1464	Standard Specification for Bent Glass
ISO 11485	Glass in Building – Curved glass



A.8 Aluminium and Framework

Reference	Standard
AS/NZS 1664	Aluminium Structures
AS/NZS 1866	Aluminium and aluminium alloys – Extruded rod, bar, solid and hollow shapes
AS 1391	Metallic materials – Tensile testing at ambient temperature
AAMA CWM	Curtain Wall Manual
AAMA TIR-A8	Structural performance of composite thermal barrier framing systems
AAMA TIR-A9	Design Guide for Metal Cladding Fasteners
AAMA 505	Dry shrinkage and composite performance thermal cycling test procedure
ANSI/AWS C5.4	Recommended Practices For Stud Welding
ASTM B221M	Standard Specification for Aluminium and Aluminium-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM E1251	Standard Test Method for Analysis of Aluminium and Aluminium Alloys by Spark Atomic Emission Spectrometry
ASTM E3061	Standard Test Methods for Analysis of Aluminium and Aluminium Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
BS EN 755	Aluminium And Aluminium Alloys. Extruded Rod/Bar, Tube And Profiles.
BS EN 1999	Eurocode 9. Design Of Aluminium Structures.
BS EN 485	Aluminium And Aluminium Alloys. Sheet Strip And Plate.
BS EN 515	Aluminium And Aluminium Alloys. Wrought Products. Temper Designations.
BS EN 573	Aluminium And Aluminium Alloys. Chemical Composition And Form Of Wrought Products.
BS EN 14024	Metal profiles with thermal barrier. Mechanical performance. Requirements, proof and tests for assessment.
CWCT	Guide To The Design Of Thermally Improved Glazing Frames

A.9 Concrete & Masonry

Reference	Standard
AS 3600	Concrete Structures
AS 3700	Masonry Structures
AS 3850	Tilt-up concrete construction

A.10 Anchor Fixings

Reference	Standard
AS 5216	Design of post-installed and cast-in fastenings in concrete
ASTM E488	Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements
BS 5080	Structural Fixings in Concrete and Masonry. Method of Test for Tensile Loading
prEN 1992-4	Eurocode 2: Design of concrete structures – Part 4: Design of fastenings for use in concrete



EOTA Tech Report 020	Evaluation of anchorages in concrete concerning resistance to fire
EOTA Tech Report 045	Design of metal anchors for use in concrete under seismic actions

A.11 Steel

Reference	Standard
AISC	Standardised Steel Connections
AS/NZS 1111	ISO Metric Hexagon Bolts And Screws
AS/NZS 1112	ISO Metric Hexagon Nuts
AS 1237	Plain Washers For Metric Bolts, Screws And Nuts For General Purposes.
AS 1252	High Strength Steel Bolts With Associated Nuts And Washers For Structural Engineering.
AS 4100	Steel Structures
AS/NZS 4600	Cold Formed Steel Structures
AS 3566	Self-drilling Screws for the Building and Construction Industries
AS/NZS 3678	Structural steel - Hot-Rolled Plates, Floor Plates and Slabs
AS/NZS 3679	Structural steel (including Parts 1 and 2)
AS/NZS 1594	Hot-Rolled Steel Flat Products
AS/NZS 1365	Tolerances For Flat-Rolled Steel Products
AS/NZS 1554	Structural Steel Welding
AS 1627	Metal finishing - Preparation and pre-treatment of surfaces
AS/NZS 2728	Prefinished/ Prepainted Sheet Metal Products For Interior/Exterior Building Applications- Performance Requirements
ISO 8501	Preparation of steel substrates before application of paints and related products – Visual assessment of surface cleanliness
ISO 8504	Preparation of steel substrates before application of paints and related products – Surface preparation methods
SSPC-SP 1	Solvent Cleaning
SSPC-SP 10	Near-White Metal Blast Cleaning (NACE No. 2)

A.12 Stainless Steel

Reference	Standard
AS/NZS 4673	Cold-formed Stainless Steel Structures
BS EN ISO 9445	Continuously Cold-rolled Steel Narrow Strip, Wide Strip, Plate/sheet and Cut Lengths. Tolerances on Dimensions and Form.
BS EN 10095	Heat Resisting Steels and Nickel Alloys.
BS EN ISO 3506	Mechanical Properties of Corrosion Resistant Stainless Steel Fasteners
BS EN 1011.3	Welding - Recommendations for Welding of Metallic Materials - Arc Welding of Stainless Steels
SCI P291	Structural Design of Stainless Steel



A.13 Zinc

Reference	Standard
BS 5625	Specification of purchasing requirements and methods of test for fluxes for soft soldering
BS EN 1179	Zinc and Zinc Alloys. Primary Zinc
BS EN ISO 9455	Soft soldering fluxes. Test methods. Flux efficacy test, wetting balance method
BS EN 29453	Soft solder alloys – Chemical Compositions and forms
BS EN 29454	Soft soldering fluxes. Classification and requirements. Classification, labelling and packaging
BS EN 29455	Soft soldering fluxes. Test methods. Determination of zinc content.

A.14 Timber

Reference	Standard
AS/NZS 1080	Timber – Methods of test
AS/NZS 1148	Timber – nomenclature – Australian, New Zealand and imported species
AS/NZS 1720.1	Timber Structures – design methods
AS/NZS 1720.1	Timber Structures – timber properties
AS/NZS 4787	Timber – Assessment of drying quality
AS 2269.0	Plywood – Structural Specification
AS 2754.2	Adhesives for timber and timber products
AS 2098.1	Methods of test for veneer and plywood – Moisture content of veneer and plywood
AS 2098.11	Methods of test for veneer and plywood – Determination of formaldehyde emissions for plywood.

A.15 Glazing Framework

Reference	Standard
AAMA CWM	Curtain Wall Manual
AAMA TIR-A8	Structural performance of composite thermal barrier framing systems
AAMA 505	Dry shrinkage and composite performance thermal cycling test procedure
BS EN 14024	Metal profiles with thermal barrier. Mechanical performance. Requirements, proof and tests for assessment.
CWCT	Guide To The Design Of Thermally Improved Glazing Frames

A.16 Sealant

Reference	Standard
AAMA 800	Voluntary Specifications and Test Methods for Sealants
ASTM C794	Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants
ASTM C920	Standard Specification For Elastomeric Joint Sealants
ASTM C1087	Standard Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems



ASTM C1135	Standard Test Method For Determining Tensile Adhesion Properties Of Structural Sealants
ASTM C1139	Standard Guide for Use of Joint Sealants
ASTM C1184	Standard Specification for Structural Silicone Sealants
ASTM C794	Standard Test Method for Adhesion – in-peel of Elastomeric Joint Sealants
ASTM D2203	Standard Test Method for Staining from Sealants
BS EN ISO 11600	Building Construction - Jointing Products- Classification and Requirements for Sealants
CIRIA	Manual Of Good Practice In Sealant Application. Special Publication 80
CIRIA FR/ CP/39	Sealant Joints In The External Envelope Of Buildings: A Guide On Design, Specification And Construction.
GANAL	Sealant Manual

A.17 Gaskets

Reference	Standard
ASTM C509	Standard Specification For Elastomeric Cellular Preformed Gasket And Sealing Material
ASTM C864	Standard Specification For Dense Elastomeric Compression Seal Gaskets, Setting Blocks And Spacers.
ASTM C1166	Standard Test Method for Flame Propagation of Dense and Cellular Elastomeric Gaskets and Accessories
ASTM D297	Standard Test Method For Rubber Products - Chemical Analysis.
ASTM D395	Standard Test Method For Rubber Property - Compression Set.
ASTM D412	Standard Test Method For Vulcanized Rubber And Thermoplastic Elastomers- Tension.
ASTM D573	Standard Test Method for Rubber – Deterioration in an Air Oven
ASTM D624	Standard Test Method For Tear Strength Of Conventional Vulcanized Rubber And Thermoplastic Elastomers.
ASTM D746	Standard Test Method For Brittleness Temperature Of Plastics And Elastomers By Impact.
ASTM D865	Standard Test Method For Rubber - Deterioration By Drying In Air (Test Tube Enclosure).
ASTM D925	Standard Test Method for Rubber Property – Staining of Surfaces (Contact, Migration, and Diffusion)
ASTM D1149	Standard Test Methods For Rubber Deterioration – Cracking In An Ozone Controlled Environment.
ASTM D2240	Standard Test Method For Rubber Property - Durometer Hardness.
ASTM D2287	Specification For Non-Rigid Vinyl Chloride Polymer And Copolymer Molding And Extrusion Compounds.
BS 4255	Rubber Used In Preformed Gaskets For Weather Exclusion From Buildings. Specification For Non-Cellular Gaskets.



A.18 Coatings

Reference	Standard
AS 1214	Hot-dip galvanized coatings on threaded fasteners (ISO metric coarse thread series)
AS 1231	Aluminium and aluminium alloys – Anodic oxidation coatings
AS 1580	Paints and Related Materials – Methods of Test
AS 3715	Metal Finishing – Thermoset Powder Coating For Architectural Applications Of Aluminium And Aluminium Alloys
AS 3750	Paints for Steel Structures
AS 3894	Site Testing of Protective Coatings
AS/NZS 4506	Metal finishing – Thermoset powder coatings
AS/NZS 4680	Hot-dip galvanized (zinc) coatings on fabricated ferrous articles
AAMA 2604	Voluntary Specification, Performance Requirements And Test Procedures For High Performance Organic Coatings On Aluminium Extrusions And Panels
AAMA 2605	Voluntary Specification, Performance Requirements And Test Procedures For Superior Performing Organic Coatings On Aluminium Extrusions And Panels
AAMA 609 & 610	Cleaning And Maintenance Guide For Architecturally Finished Aluminium
AAMA 611	Voluntary Specification For Anodised Architectural Aluminium
AAMA 621	Voluntary Specifications For High Performance Organic Coatings On Coil Coated Architectural Aluminium Substrates
BS EN 13438	Paints and varnishes. Powder organic coatings for hot dip galvanised or sherardised steel products for construction purposes
BS 6338	Specification for Chromate Conversion Coatings on Electroplated Zinc and Cadmium Coatings
ABCB	Handbook: Durability in Buildings including Plumbing Installations
GAA	Galvanizers Association of Australia: 'Hot Dip Galvanizing'
NAAMM	Metal Finishes Manual
NZBC B2/VM1 & AS1	Acceptable Solutions and Verification Methods - B2 Durability

A.19 Cladding Systems and Waterproofing

Reference	Standard
AS 1562.1	Design and installation of sheet roof and wall cladding Part 1: Metal
AS 1562.3	Design and installation of sheet roof and wall cladding Part 3: Plastic
AS/NZS 4200.1	Pliable building membranes and underlays Part 1: Materials
AS/NZS 4200.2	Pliable building membranes and underlays Part 2: Installation requirements
AS 4654	Waterproofing membranes for external above ground use

A.20 Access & Maintenance

Reference	Standard
AS/NZS 1576	Scaffolding



AS 1657	Fixed platforms, walkways, stairways and ladders - Design, Construction and Installation
AS/NZS 1891	Industrial Fall Arrest Systems & Devices
AS/NZS 1892	Portable Ladders
AS/NZS 4576	Guidelines for Scaffolding

A.21 Testing

Reference	Standard
AS 2047	Windows in buildings – Selection and Installation
AS 4420	Windows – Methods of Test
AS/NZS 4284	Testing of building facades
AS/NZS 4347	Damp-proof Courses and Flashing – Methods of Test
AS 5203	Protection of openable windows/fall prevention – Test sequence and compliance method
AAMA 501.1	Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure
AAMA 501.2	Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls, and Sloped Glazing Systems
AAMA 501.8	Standard Test Method for Determination of resistance to Human Impact of Window Systems Intended for Use in Psychiatric Applications
AAMA 502-08	Voluntary Specification for Field Testing of Newly Installed Fenestration Products
AAMA 503-08	Voluntary Specification for Field Testing of Newly Installed Shopfronts, Curtain Walls and Sloped Glazing Systems
AAMA 508-14	Voluntary Test Method and Specification for Pressure Equalized Rain Screen Wall Cladding Systems
AAMA 509-14	Voluntary Test and Classification Method for Drained and Back Ventilated Rain Screen Wall Cladding Systems
AAMA 511-08	Voluntary Guideline for Forensic Water Penetration Testing of Fenestration Products

A.22 Thermal Calculations

Reference	Standard
BS EN 12524	Building materials and products. Hygrothermal properties. Tabulated design values.
BS EN ISO 10211	Thermal bridges in building construction. Heat flows and surface temperatures. Detailed calculations.
DIN 4108	Thermal Insulation And Energy Economy in Buildings – Part 4: Hygrothermal Design Values
BS EN ISO 6946	Building components and building elements – Thermal resistance and thermal transmittance – Calculation methods
BS EN ISO 10077-1	Thermal performance of windows, doors and shutters – Calculation of thermal transmittance – Part 1: General
BS EN ISO 10077-2	Thermal performance of windows, doors and shutters – Calculation of thermal transmittance – Part 2: Numerical method for frames



A.23 Hardware

Reference	Standard
BS EN 1527	Building hardware – Hardware for sliding doors and folding doors
BS EN 13126	Building hardware. Requirements and test methods for windows and door height windows

A.24 Other

Reference	Standard
AS 1428	Design for access and mobility
AS 4964	Method for the qualitative identification of asbestos in bulk samples



B **Appendix B – Contractor Submissions**

This table (or equivalent) is to be used by the Head Contractor and/or the Specialist Façade Contractor(s) to record and track the submissions required by this Specification. It covers all phases, including design, fabrication and installation.

Provide periodic (monthly) updates to Inhabit of Contractor Submissions status

B.1 **Design and verification submissions**

Spec Clause	Submission – Design and Verification	Timeframe	Initial Submission	Final Submission	Approval Date
3.3.4	Safety and Human Impact				
3.10.1	Maintenance				
2.2.2	Shop drawings				
2.2.3	Prototypes				
2.2.4	Maintenance and Operating Manual				
2.2.5	As Built drawings				
2.4.2	Façade Contractor’s Professional Engineer	Prior to commencement of design			

Spec Clause	Submission – Design and Verification	Timeframe	Initial Submission	Final Submission	Approval Date
1.1.1	Sealant Supplier, Proprietary Louvres Proprietary System	Computations and Shop Drawings			
2.5	Quality Assurance	Prior to commencement of design			
2.6	Certification Design certification	Prior to commencement of fabrication			
2.8	Delivery, Handling & Storage	Prior to commencement of fabrication			
2.9	Project Conditions Survey and Set-Out	Prior to commencement of fabrication			
2.10	Performance Testing	Prior to commencement of fabrication			
2.10.5	Manufacturer's Standard Test Reports	Prior to commencement of fabrication			
3.13	Safety in Design	Computations and Shop Drawings			
3.3.2	Compatible Materials	Prior to commencement of fabrication			
3.3.5	Panel Replacement	Prior to commencement of fabrication			
1.1		Prior to commencement of fabrication			
3.8.6	Thermal Performance	Prior to commencement of fabrication			

Spec Clause	Submission – Design and Verification	Timeframe	Initial Submission	Final Submission	Approval Date
1.1.1	Logistics of glass processing and programme	With tender			
4.2	Elastomeric Extrusions, Gaskets, Seals, Setting Blocks & Accessories Details Compatibility with adjacent substrates	Prior to commencement of fabrication			
TBC	Façade and Interface details for: NSW buildings Class 2 buildings. Façade and Interface details to be stamped and logged on NSW Portal	TBC			
TBC	FP1.4 interface and verification detail	TBC			

B.2 Fabrication submissions

Spec Clause	Submission – Fabrication	Timeframe	Initial Submission	Final Submission	Approval Date
2.5	Quality Assurance – Quality Plans	Prior to commencement of fabrication			
2.10.4	Structural Silicone Testing				
3.11.2	Visual Mock-Up (VMU) / Prototype				
3.11.2.1	Visual Mock-Up (VMU) – Glass Samples				

Spec Clause	Submission – Fabrication	Timeframe	Initial Submission	Final Submission	Approval Date
4.0	Materials Certification ^ ^	Prior to commencement of fabrication			
4.1	Glass ISO Accreditation of Glass Manufacturer	Prior to ordering of glass			
4.1.1	Glass Thermal Properties NFRC Conditions	Prior to ordering of glass			
4.1.2	Glass Quality Requirements Production and test records to ASTM 1036 Manufacturer's data related to nickel sulphide control	During glass manufacture			
4.1.4	Annealed Glass Production and test records to ASTM 1036 Thermal stress analysis	Prior to ordering of glass			
4.1.5	Heat Strengthened and Toughened Glass Production and test records of base material to ASTM 1036 Non-destructive testing data to confirm surface compressive stress	During glass manufacture and fabrication			
4.1.5.1	Toughened Glass Heat soaking methodology	Prior to ordering of glass			
4.1.6	Laminated Glass Production and test records to ASTM C1172, AS/NZS 2208 Compatibility with proposed sealants	Prior to ordering of glass and during glass manufacture			
4.1.7.4	Insulating Glass Units Quality Records Production records	Prior to ordering of glass and during glass manufacture			

Spec Clause	Submission – Fabrication	Timeframe	Initial Submission	Final Submission	Approval Date
4.1.7.5	Insulating Glass Units Testing Test records to ASTM E773 and ASTM E774	During glass manufacture			
4.1.8	Low-E Coated Glass Visual quality control acceptance criteria	Prior to ordering of glass			
4.1.9.1	Heat Treated Glass Manufacture QA Documents – checking of quench patterns	During glass manufacture			
4.1.10	Ceramic Enamelled (Fritted) Heat Treated Glass Direct contact with laminating interlayer and/or Applied to low-E glass - Confirmation from Glass Supplier	Prior to ordering of glass			
4.2.1	Elastomeric Extrusions, Gaskets, Seals, Setting Blocks & Accessories Production and test records	During glass manufacture			
4.2.2	Elastomeric Extrusions, Gaskets, Seals, Setting Blocks & Accessories Warranties	Prior to ordering of materials			
4.6.4.1	Steel Coatings Information about Steel Coating Operator and Coating Facilities Proposed coating, surface treatment, method of application, equipment etc	Prior to commencement of coating work.			
4.6.4.11	Steel Coatings Inspection and Testing	During coating work			
4.6.4.17	Steel Coatings Trial Application – Test Area, Panel	Prior to commencement of coating work.			
4.6.4.18	Steel Coatings Future maintenance requirements	Prior to commencement of coating work			

Spec Clause	Submission – Fabrication	Timeframe	Initial Submission	Final Submission	Approval Date
4.7	Aluminium Details of extrusions, alloy types	Prior to ordering of materials			
4.7	Aluminium Aluminium COCs or test reports	Prior to fabrication			
4.8.8.3	Aluminium Quality records and test reports of aluminium coatings	During fabrication			
4.8.8.4	Aluminium Cleaning, Maintenance Repair procedures for Coatings	During fabrication			
4.8.8.5	Aluminium Powder Coat Finishes Details of preparation and pretreatment Details of Qualicoat applicator	Prior to powder coating work			
4.8.8.6	Aluminium Anodise Finishes Details of preparation and pretreatment Details of Anodiser	Prior to anodising work			
4.10	Timber Chain of Custody COC Audit of listed materials	During fabrication			
4.10.4.1	Timber Samples of timber species	Prior to ordering of materials			
4.10.4.2	Timber Samples Samples demonstrating variation in appearance	Prior to ordering of materials			
4.10.4.4	Timber Benchmark samples	Prior to ordering of materials			
5.1.1.1	Aluminum Fixings Proposed Testing Programme	During fabrication			

Spec Clause	Submission – Fabrication	Timeframe	Initial Submission	Final Submission	Approval Date
5.1.13.1	Louvres and Grillages Submit verification of compliance with free-air movement and maximum pressure drop requirements.	Prior to ordering of materials			
TBC	FP1.4 system off-site verification register completed	TBC			

B.3 Installation submissions

Spec Clause	Submission – Installation	Timeframe	Initial Submission	Final Submission	Approval Date
2.5	Quality Assurance – Quality Plans	Prior to commencement of installation			
2.6	Certification Acceptance of substrates	Prior to commencement of installation			
2.10.3	Field Testing	During installation			
4.10	Timber Audit of listed materials	During installation			
5.1.13.7	Louvres and Grillages Demonstrate louvres are demountable and can be easily removed and reinstalled without damaging surrounding panels.	During installation			
5.3	Handling Provide all required packing, wrapping and handling techniques to ensure that all panels are installed and handed over without any damage to the finished panel faces.	During installation			

Spec Clause	Submission – Installation	Timeframe	Initial Submission	Final Submission	Approval Date
5.4.2	Acceptance of substrates Submit certification of acceptance of substrates and conditions, including the provision of reference lines and marks, and making specific reference to the type, number and location of the built-in anchors provided in the base-structure for the installation.	Prior to commencement of installation			
5.4.9	Hardware Complete site ITP checks for each operable façade component Take and maintain formal records of these checks and adjustments.	During installation			
5.4.12	Acceptance of sub-framework Submit certificate by licensed surveyor to confirm that base framing is in the correct locations and to the required tolerances prior to commencement of installation.	During installation			
TBC	System installation and interfaces benchmarked by the superintendent (inspection)	TBC			
TBC	FP1.4 system on-site verification register completed.	TBC			
TBC	On-site protection plan (in completion of handling cl 5.3)	TBC			
TBC	Façade defects register provided	TBC			
TBC	Combustibility	TBC			